MILITARY SPECIFICATION
RADIO TELETYPEWRITER SET AN/GRC-142( )
IN S-318 SHELTER

1. SCOPE

1.1 This specification covers a shelter installed radio communications system Radio Teletypewriter Set AN/GRC-142( ) which provides one way reversible radio teletypewriter, amplitude-modulated voice, single side band voice and continuous wave CW communications in the high frequency band. Facilities are also provided to allow the ready installation and operation of tactical electronic teletypewriter security equipment when secure radio teletypewriter operation is required. Provisions are made for operation of the AN/GRC-142( ) from either 28 volts D.C. or 115 volts 1 phase 50/60 Hz AC primary power sources. Provisions are made for installation of auxiliary equipment to permit full duplex voice and radio teletypewriter operation, when required.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571	Solder: Lead alloy, tin lead alloy and Tin Alloy; Flux Cored Ribbon and Wire, and solid form

MILITARY

MIL-E-1	Electron Tubes and crystal rectifiers
MIL-C-17	Cables, radio frequency, coaxial, dual coaxial, twin conductor and twin lead
MIL-P-116	Preservation, methods of
MIL-V-173	Varnish, moisture and fungus resistant, for the treatment of communications, electronic and associated electrical equipment
MIL-T-713	Twine & Tape, lacing and tying (for use in electrical and electronic equipment)
MIL-C-3432	Cable, flexing and extra flexible 300 & 600 volts
MIL-C-3885	Cable assemblies and cord assemblies, electrical (for use in electronic, communication and associated electrical equipment)
MIL-W-6858	Welding, resistance, aluminum, magnesium, nonhardening steels or alloys, nickel alloys, heat-resisting alloys and titanium alloys, spot & seam
MIL-B-7883	Brazing of steels, copper, copper alloys and nickel alloys
MIL-T-7928	Terminal lugs crimp style
MIL-C-10544	Connector, plug & receptacle (electrical, audio waterproof ten contact, polarized)

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SPECIFICATIONS (cont)

MILITARY

MIL-M-10578	Metal conditioner and rust remover (phosphoric acid base)
MIL-M-10953	Manual for literature (Tech Man and Tech Bulletins and changes) Signal Corps Equipment
MIL-P-11268	Parts, materials, and processes used in electronic communication equipment
MIL-M-13231	Marking of electronic items
MIL-F-14072	Finishes for ground signal equipment
MIL-F-14256	Flux, soldering, liquid (rosin base)
MIL-P-17257	Padlock dial combination
MIL-S-19500	Semiconductor devices, general specification for
MIL-N-25027	Nut, self-locking 250°F., 500°F, and 800°F
MIL-C-3702	Cable, Power, Electrical; Ignition, High-Tension
MIL-C-26482	Connectors, electrical, circular miniature, quick disconnect
MIL-I-46058	Coatings, conformal, insulating compound electrical (for coating painted circuit assemblies)
MIL-H-52204	Retainer assembly - sliding
MIL-P-55110	Printing wiring boards
MIL-M-55234	Standing wave ratio meter power ME-165( )/G
MIL-S-55429	Shelter, electrical equipment S-318( )/G
MIL-S-55505	Shelter, electrical equipment (containing equipment) Packaging and Packing of

STANDARDS

FEDERAL

FED-STD-222	Radiation Standard for communications and other information processing equipment
FED-STD-595	Colors

STANDARDS

MILITARY

MIL-STD-22	Welded-Joint Designs
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment
MIL-STD-200	Electron Tubes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements
MIL-STD-462	Electromagnetic Interference, Characteristics, Measurements
MIL-STD-681	Identification Coding and Application of Hookup Wire
MIL-STD-701	Preferred and Guidance List of Transistors
MIL-STD-810	Environmental Test Methods
MS-33540	Safety Wiring, General Practices for

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STANDARDS (cont)

MILITARY

MS-75027 Cable Cord, Standard Lengths and Standard Tolerances, for Communication, Power and Radio Frequency Cable and Cord Assemblies

MISCELLANEOUS STANDARDS AND SPECIFICATIONS

SM-D-603491 Device Low Level Signalling TT-523A/GGC
SCL-2101 Reports, Equipment Compliance

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439 List of Accessories for Package Tester
SC-B-19491 Clamp, Loop Antenna Rope
SC-B-61578 Grounding to Chassis
SC-C-4500 Impedance curves for 15 foot vehicle Mounted Whip Antenna

SM-D-602939 Installation of AN/GRC-142 components in Shelter S-318/G
SM-C-603830 Power Supply (Modified)
SM-D-602945 Switch Box SA-1554/GRC-142
SC-DL-502745 Cross Bar Assembly
SM-D-14158 Rod, Ground
SM-C-603093 Hammer Sledge
SM-D-603394 Tarpaulin
SM-C-603270 Marker Cable
SC-DL-49678 Mast Base AB-652/GR
SC-DL-349717 Standing Wave Ratio Power Meter ME-165( )/G
SC-DL-603308 Remote Control C-7279/GRC-142

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DRAWINGS (cont)

ELECTRONICS COMMAND

SC-DL-603309	Entrance Box Assembly DC
SC-DL-603310	Entrance Box Assembly AC
SC-DL-603311	Roadside Storage Box
SC-DL-603312	Roadside Storage Box
SC-DL-603313	Box Telephone
SC-DL-603314	Switch Box SA-1554/GRC-142
SC-DL-603315	Panel, Power Distribution SB-3018/GRC-142
SC-DL-603316	Switch Box SA-1555/GRC-142
SC-DL-603317	Interconnecting Box, J-2728/GRC-142
SC-DL-602939	Base and Hardware for Mounting Power Supply PP-4763/GRC

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications.

The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Standards Association

Z24.10-1953 Octave Band Filter Set for the Analysis of Noise and Other Sounds

(Copies can be obtained from United States of America Standards Institute, 10 East 40th Street, New York City, New York)

CONSOLIDATED CLASSIFICATION COMMITTEE
Consolidated Freight Classification
Uniform Freight Classification

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(Applications for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Ill.).

NATIONAL BUREAU OF STANDARDS
Handbook H-28 Screw-Thread Standards for Federal Services.

POST OFFICE DEPARTMENT
Parcel Post Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

AMERICAN TRUCKING ASSOCIATION, INC.
National Motor Freight Classification.

(Application for copies should be addressed to the National Classification Board, 1424 15th Street NW, Washington 6, D.C.).

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3. REQUIREMENTS

3.1 Description: Radio Teletypewriter Set AN/GRC-142( ) is a tactical high frequency radio telephone and radio teletypewriter communications facility mounted in a shelter and capable of operation at halt, as well as in motion, while being transported by an appropriate army vehicle. This specification covers installation of the AN/GRC-142( ) in a 3/4 ton shelter, type S-318/G. Requirements peculiar to the S-318/G are identified in the paragraphs to follow.

3.1.1 General Requirements: The contractor shall furnish and install specified items of contractor-furnished equipment in and on the shelter. The contractor shall also install specified items of Government furnished equipment in and on the shelter and shall interconnect all electrical, electronic, heating and ventilating equipment to form an operable assembly capable of meeting the performance requirements and installation criteria as given in the following paragraphs. The contractor shall furnish and install all hardware, racks, brackets, cabling and other miscellaneous items not specifically covered by this specification to provide a complete and fully operational AN/GRC-142( ).

3.2 Procurement Reference Model: A procurement reference model of the shelter S-318/G configuration of Radio Teletypewriter Set AN/GRC-142( ) will be made available for inspection by prospective bidders and will be loaned to the successful bidder after award of contract. Unless otherwise specified herein, or in the Invitation for Bids, the physical location of components in and on the S-318/G shelter as called for in the bid request and contract, shall conform to the procurement reference model.

3.2.1 Operational and Maintenance Facilities: The operational and maintenance facilities and the electrical, mechanical and environmental performance of the equipment shall be equal to or better than the reference model except where specifically changed by the specification and/or list of exceptions to the model. In all cases of conflict between the reference model and the specification, the specification shall govern. It shall be incumbent upon the contractor to inspect and test the procurement model to determine its configuration, facilities, capabilities and performance. The contractor shall advise the Contracting Officer's Technical Representative of all conflicts between the model and specification and shall submit for approval all changes from the model to be incorporated into the production equipment as a result of:

a. Production fabrication methods and techniques.

b. The list of exceptions to the model.

c. Specification requirements.

d. Cost reduction value engineering.

e. Product improvement and/or simplification.

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3.2.2 Drawings: Drawings covering design, construction, installation, assembly and interconnection of various items in the S-318/G configuration will be made available at time of solicitation for bids. In all cases of conflict between such drawings and this specification, this specification shall govern. The contractor shall advise the COTR of all conflicts noted between Government furnished drawings and this specification and request resolution of the conflicts.

3.2.3 Human Engineering: The contractor shall report on his effort with regard to human engineering.

3.2.4 Tempest: The Plan shall include complete discussion and details of tests for meeting the requirements of Federal Standard 222. This Plan shall be submitted under separate cover, in accordance with appropriate security regulations.

3.2.5 Electromagnetic Compatibility: The interference control plan specified in 4.2 of MIL-STD-461 shall be submitted.

3.3 First Article Samples: The contractor shall furnish First Article samples of the AN/GRC-142( ) for approval in quantities as specified in the Invitation for Bids and contract.

3.3.1 Reference Standards: After the First Article samples have been approved, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.3.2 Changes: After approval of the First Article samples, no changes shall be made without Contracting Officer's Technical Representative (COTR) approval. Upon generation of a proposed change, the contractor shall submit to the Contracting Officer a complete plan fully describing the change, with substantiating test data, and intended method(s) of implementing the proposed change. No action by the contractor to enforce a change shall be made without prior approval of the Engineering Change Proposal (ECP).

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3.4 Form and Arrangement of Equipment: The form and arrangement of all the equipment on and in the S-318/G Shelter shall conform to the procurement reference model and drawings except as otherwise specified in the contract. The fabricating techniques employed in the assembly of the equipment shall be such that the equipment will be capable of withstanding transportation over all types of terrain and conditions of temperature and humidity normally encountered in military use. This involves stresses resulting from various types of shock, vibration, depot stacking, high altitude, air delivery (rotary and fixed wing types of aircraft) and parachute drop. The equipment shall be capable of being handled or transported without breakage, permanent deformation or loosening of threaded or other parts. It shall be capable of operating in all weather conditions encountered in temperate, arctic, and tropical climates unless otherwise specified herein.

3.5 Weight: The total weight of the shelter with all units of equipment installed shall not exceed the following: S-318/G configuration: 1,580 pounds.

Every effort shall be made by the contractor to minimize the weight of all units, racks, brackets, tables, etc. to an absolute minimum consistent with the service conditions requirements of this specification. Center of gravity shall be such as to permit the fully equipped shelter to meet the requirements for orientation as stated in 3.41.11 and vehicular transport as stated in 3.41.10.

3.6 Operational Characteristics: Radio Teletypewriter Set AN/GRC-142( ) shall provide the following transmission and reception capabilities throughout the frequency range 2.0 to 29.999 MHZ "in motion" and "at halt" on a simplex (one way reversible) basis:

a. 85 HZ and 850 HZ frequency shift keyed radio teletypewriter operation.

b. Compatible AM and SSB voice operation.

c. Simultaneous 85 HZ frequency shift keyed radio teletypewriter and SSB voice operation.

d. 85 HZ frequency diversity, frequency shift keyed radio teletypewriter operation.

e. CW operation.

f. Remote operation

In all modes involving radio teletypewriter operation, not more than one error per twenty lines of "Fox" copy will be allowed during tests prescribed herein. Specified audio frequency and radio frequency characteristics of Radio Set AN/GRC-106 shall be unaffected by integration of this radio set into the AN/GRC-142( ) system.

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3.6.1 Facilities for Duplex Operation: Facilities including switching, cabling and mounting provisions shall be included to enable operation of the radio teletypewriter set on a full duplex basis when appropriate auxilliary equipment (not part of AN/GRC-142( )) is added. Full duplex operation while in motion is not required.

3.6.2 Remote Operation: When set up at a semi-fixed location, the AN/GRC-142 shall provide for operation from a remote site at distances up to one mile, in operating modes listed in 3.6a, 3.6b, 3.6c, 3.6d and 3.6e on a simplex (one way reversible) basis. Full duplex operation shall be possible at distances up to one mile, in operating modes listed in 3.6a, 3.6d and 3.6e, when appropriate ancilliary equipment (not part of AN/GRC-142( )) is added. The AN/GRC-142( ) shall be capable of switching from full duplex operation to one way reversible operation and using the "duplex receive" local and remote teletypewriters (not part of AN/GRC-142( )) in a "pony" teletypewriter circuit to the remote site. Control Group AN/GRA-6 (part of AN/GRC-142( )) shall be used for remote voice transmission and reception as well as for the "send-receive" remote control function in radio teletypewriter operation. Telephones TA-312/PT (two provided as part of AN/GRC-142( )) shall be used to provide a voice "order wire" circuit between the local and remote sites. Telegraph Key KY-116A/U (part of AN/GRC-142 ( )) shall be used for local or remote transmission of CW signals. Headset H-227/U (part of AN/GRC-142( )) shall be used for remote reception of CW signals. Facilities shall be provided for interconnecting the AN/GRC-142( ) shelter and the remote site using a remote box (part of AN/GRC-142( )) and a maximum of five pairs of field wire as follows:

a. Pair Number 1: TA-312/PT "order wire" telephone.

b. Pair Number 2: Control Group AN/GRA-6.

c. Pair Number 3: One way reversible (or DX SEND) teletypewriter DC loop circuit.

d. Pair Number 4: Full duplex receive (or "pony" circuit) teletypewriter DC loop circuit.

e. Pair Number 5: Remote CW.

3.7 Facilities for Secure Radio Teletypewriter Operation: Physical security measures, equipment mounting facilities and electrical signal and power cabling shall be provided to enable use of appropriate radio teletypewriter security equipment. Details are given in the applicable paragraphs of this specification.

3.8 Teletypewriter System Requirements: The operating, control and auxilliary components which comprise Radio Teletypewriter Set AN/GRC-142( ), and the ancilliary items (not part of AN/GRC-142( )) when installed, shall perform the following functions in both clear text and secure modes:

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3.8.1 Simplex Operation (one way reversible): Simplex (one way reversible) operation using Radio Set AN/GRC-106, Radio Teletype Modem MD-522A/GRC, Teletypewriter TT-98/FG, and Reperforator-Transmitter TT-76B/GGC shall provide the following operating capabilities:

a. "On line" page copy reception of radio teletype signals on the TT-98/FG and at the same time, independent "off line" preparation of punched tape using the reperforator and keyboard or distributor of the TT-76B/GGC.

b. "On line" reception of radio teletype signals on the TT-98/FG page printer and/or reperforator of the TT-76B/GGC.

c. "On line" transmission of teletype signals using the keyboard of the TT-98/FG and/or TT-76B/GGC with monitoring of transmitted message on the page printer of the TT-98/FG.

d. "On line" transmission of teletype signals using the TT-76B/GGC teletype distributor and a prepared tape, with monitoring of transmitted message on the TT-98/FG page printer and simultaneous "off line" preparation of a tape using the TT-76B/GGC keyboard and the reperforator.

3.8.2 Duplex Operation: Full duplex operation using Radio Set AN/GRC-106, auxiliary Receiver-Transmitter RT-662/GRC *, Radio Teletypewriter Modem MD-522A/GRC, Teletypewriter TT-98/FG, Reperforator-Transmitter TT-76B/GGC, and auxiliary antenna equipment *, auxiliary inverter *, auxiliary TT-98/FG *, and Tuning Unit, R.F. TN-510/GRC * shall provide the following operational capabilities: (Items marked with an asterisk are not part of AN/GRC-142( )):

a. "On line" reception of radio teletype signals on the auxiliary TT-98/FG page printer * and/or TT-76B/GGC tape reperforator and simultaneous independent "on line" transmission of radio teletype signals from the keyboard of the TT-98/FG, keyboard of the TT-76B/GGC and/or tape distributor of the TT-76B/GGC, with monitoring of the transmitted message on the page printer of the TT-98/FG.

b. "On line" reception of radio teletype signals on the auxiliary TT-98/FG page printer * and simultaneous independent "on line" transmission of radio teletype signals using the TT-98/FG keyboard and/or TT-76B/GGC tape distributor and a prepared tape, with monitoring of the transmitted message on the page printer of the TT-98/FG and simultaneous independent "off line" preparation of punched tape using the keyboard and reperforator of the TT-76B/GGC.

3.8.2.1 Duplex operation of the AN/GRC-142( ) shall be possible at any frequency from 2.0 to 29.999MHZ under the following conditions:

a. Transmitted output at 400 watts PEP.

b. Transmitting antenna is a doublet or 15 foot whip. The receiving antenna is a doublet, physically separated from the transmitting antenna by a maximum distance of 150 feet.

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c. The receiver frequency may be as close to the transmitter frequency as 10% of the transmitter frequency, when using Tuning Unit, Radio Frequency, TN-510/GRC (R.F. Preselector).

d. Receiver cross modulation distortion is -26db or lower.

e. When doublet antennas are used for transmitting and receiving, the antennas shall be set up "end to end" and not facing each other "broadside". A preselector (not part of AN/GRC-142( )) shall be used to meet the requirements of c and d above. The preselector provides the necessary selectivity ahead of the receiver antenna input and is manually tuned.

3.9 System Electrical Characteristics. Reference Military Standard 454.

3.9.1 Warm-up Time: Warm-up time for operation at full specified performance will be governed by the characteristics of the Government furnished equipment installed in the shelter.

3.9.2 Primary Power Inputs:

a. 24.0 to 30.0 volts DC, 100 ampere capacity, 1.0% regulation, 1.0% ripple, with transient peaks resulting from sudden changes in load not to exceed 35.0 volts. A power source with the above characteristics shall be used when the AN/GRC-142( ) is set up for "DC" operation. In this mode of operation, the primary DC input shall provide power to the following installed components:

(1) Radio Set AN/GRC-106, consisting of Receiver-Transmitter RT-662/GRC and Amplifier RF, AM-3349/GRC-106.

(2) Modem Radio Teletypewriter, MD-522/GRC or MD-522A/GRC.

(3) Shelter lights (50 watts total).

(4) Lockout relay coil.

(5) Inverter, loaded by one each teletypewriter TT-98/FG and one each Teletypewriter Reperforator-Transmitter TT-76B/GGC.

(6) DC voltmeter.

(7) Power Panel indicator lamp.

(8) Security device (used with, but not part of AN/GRC-142( )).

(9) Exhaust fan.

(10) 28 volt convenience outlet with contact arrangement which precludes mating with standard AC parallel blade plug.

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(11) Multi-fuel heater.

(12) Auxiliary inverter, loaded by one each auxiliary teletypewriter TT-98/FG (used for full duplex operation and not part of AN/GRC-142( )).

(13) Auxiliary security device (used for full duplex operation and not part of AN/GRC-142( ).

(14) Auxiliary Receiver-Transmitter RT-662/GRC (used for full duplex operation and not part of AN/GRC-142( )).

(15) Auxiliary RT-662/GRC RF Preselector (used for full duplex operation and not part of AN/GRC-142( )). Tuning Unit, Radio Frequency TN-510/GRC.

b. 103.5 to 126.5 volts AC, single phase, 47 to 63 HZ, 25 ampere capacity. A power source with the above characteristics shall be used when the AN/GRC-142( ) is set up for "AC" operation. In this mode of operation the primary AC input shall provide power to the following installed components:

(1) Power Supply PP-4763/GRC loaded by:

a. Radio Set AN/GRC-106 consisting of Receiver-Transmitter RT-662/GRC and Amplifier, RF AM-3349/GRC-106.

b. Modem, Radio Teletypewriter, MD-522/GRC or MD-522A/GRC.

c. Shelter lights (50 watts total).

d. Lockout relay coil.

e. Exhaust fan.

f. Auxiliary Receiver-Transmitter RT-662/GRC (used for full duplex operation and not part of AN/GRC-142( )).

g. Auxiliary RT-662/GRC RF Preselector (used for full duplex operation and not part of AN/GRC-142( )).

(2) Teletypewriter TT-98/FG.

(3) Teletypewriter Reperforator-Transmitter TT-76B/GGC.

(4) Security device (used with but not part of AN/GRC-142( )).

(5) Auxiliary security device (used for full duplex operation and not part of AN/GRC-142( )).

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(6) Auxiliary Teletypewriter TT-98/FG (used for full duplex operation and not part of AN/GRC-142( )).

(7) AC voltmeter.

c. An AC/DC Power Transfer Switch shall be provided to permit operation of the equipment as described in a and b above. The transfer switch, with housing, shall be constructed, installed and interconnected into the AN/GRC-142 ( ) system in accordance with applicable paragraphs of this specification.

3.9.3 Frequency Coverage: The frequency coverage shall be 2.0 through 29.999 MHZ in 1 KHZ detented steps transmitting, and 2.0 through 29.999 MHZ in 1 KHZ detented steps with vernier tuning between detented steps while receiving.

3.9.4 RF Power Output: The installed AN/GRC-106 Radio Set shall have the following level of power output into a 50 ohm load over the frequency range of 2.0 to 29.999 MHZ for the following modes of emission when the primary input DC voltage is varied as specified in 3.9.2a: See 4.10.5.

3.9.4.1 SSB Voice: When the equipment is modulated by the standard test message of 4.10.1.6, the equipment shall be capable of a peak envelope power output of at least 400 watts. When modulated by two equal sinusoidal audio tones (1500 and 2500 HZ), the peak envelope power output shall be 400 watts +2, -1 db.

3.9.4.2 Compatible AM: The average power output of the compatible AM signal (carrier and upper side band) when modulated by the standard test message of 4.10.1.6 shall be at least 200 watts. The carrier power shall be 70 watts +2, -1 db.

3.9.4.3 CW, NSK and FSK: The average power output shall be 200 watts +2, -1 db.

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3.9.5 Hum and Noise: The hum and noise output of the installed AN/GRC-106 Radio Set when measured at the antenna terminals of the equipment shall have a value of more than 50 db below either of the two equal output frequencies corresponding to two audio input modulating tones with frequencies of 1500 and 2500 HZ when the AN/GRC-106 is operating at any power level from 100 watts PEP up to and including rated peak envelope power output at any frequency in the range 2.0 to 29.999 MHZ. This requirement shall not supersede any requirement of MIL-STD-461. See 4.10.3.

3.9.6 Transmission Continuity: The transmitted output of the radio teletypewriter set shall be continuous under conditions of vibration as experienced in the road test of 4.9.10.1. See 4.10.6.

3.9.7 CW Keying: Keying speeds up to 25 dot cycles per second shall be possible. The radiated signal shall fall within the shaded portion of Figure 1, when the AN/GRC-106 Radio Set is keyed through the AN/GRC-142( ) system from the shelter as well as from a remote site using one mile of field wire, or its equivalent. See 4.10.4.

3.9.8 Safety and Protective Devices. See 4.10.2.

3.9.8.1 Voice "Lock-Out" Provisions: Circuitry shall be incorporated in the system wiring to prevent voice modulation of the following installed components, when any radio teletypewriter security device is being used in the AN/GRC-142( ):

a. Radio Set AN/GRC-106.

b. Telephone TA-312/PT.

c. Control Group AN/GRA-6.

3.9.8.2 Shelter Grounding Facilities: A ground stud shall be incorporated on the DC Power Entrance Panel, 3.11.3.5, for connection to a ground rod, 3.34, by means of a ground braid. Reference Electronics Command Drawing SC-B-61578.

3.9.8.3 Primary Power Protection: Protective devices as listed below shall be incorporated into contractor-furnished components of the AN/GRC-142( ):

a. AC Power Entrance Panel: One (1) each circuit breaker with 25 ampere rating for all other AC service in the shelter.

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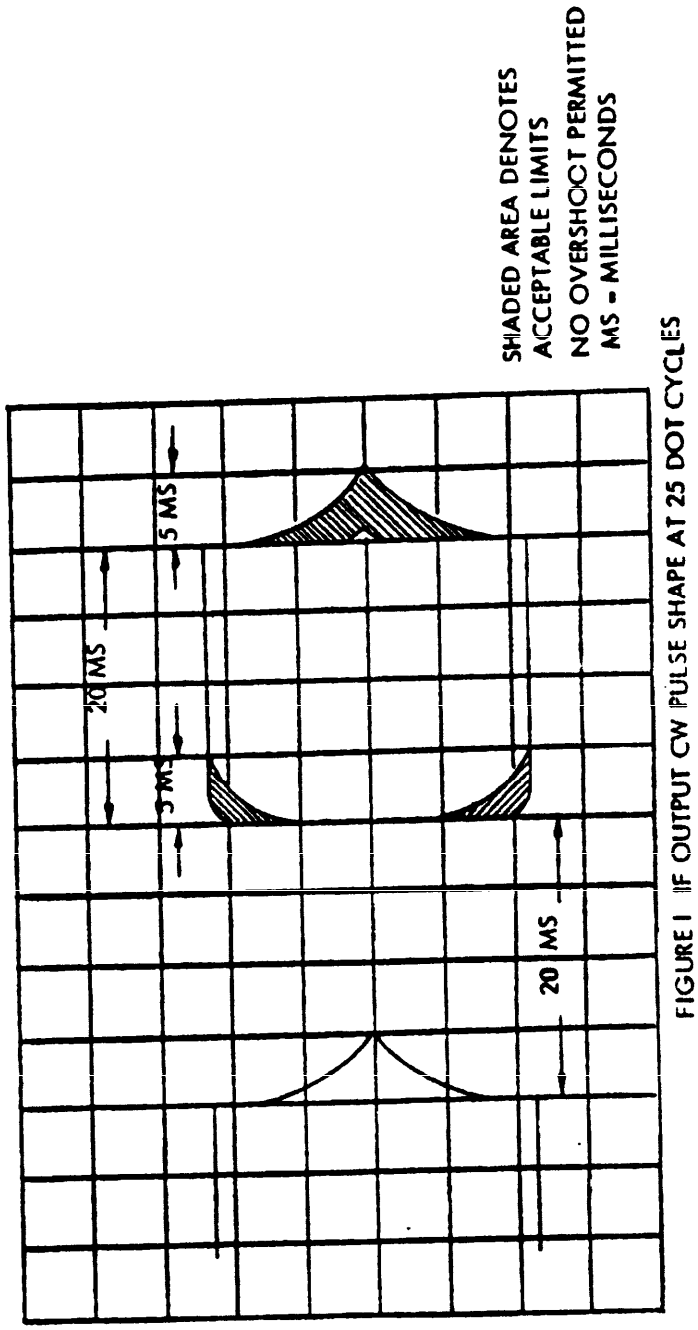


FIGURE I IF OUTPUT CW PULSE SHAPE AT 25 DOT CYCLES

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b. DC Power Entrance Panel: Lightning arrestors per paragraph 3.9.8.5.

c. Power Panel:

(1) Fuses, per paragraph 3.18.

(2) Main circuit breaker per paragraph 3.18.

(3) Lockout provisions to remove power from main and auxiliary security device DC power cables when the AC/DC Power Transfer Switch (3.9.2c) is set to AC, and remove power from main and auxiliary security device AC power cables when the AC/DC Power Transfer Switch is set to DC.

3.9.8.4 Electrical Safety: The equipment, when set up for operation, or in a servicing condition, with access covers in place, shall protect personnel from contact with potentials in excess of 30 volts to ground. All access covers, into areas where potentials in excess of 30 volts are present, shall have warning signs (DANGER HIGH VOLTAGE) affixed. Access covers into areas where high DC current capacity is present, shall have warning signs (DANGER HIGH AMPERAGE) affixed. Two (2) "DANGER HIGH VOLTAGE" warning signs shall be affixed to the outside of the shelter near the AN/GRC-106 whip antenna base. One sign shall be affixed to the front of the shelter. The second sign shall be affixed to the roadside of the shelter. A sign or stencil reading "USE ANTENNA SHEATH AT ALL TIMES" shall be affixed adjacent to the high voltage warning sign on the front of the shelter. Reference Military Standard 33540.

3.9.8.5 Lightning Protection: Lightning arrestors shall be installed in the DC power entrance panel to protect against lightning discharge on the five pairs of field wire which may be connected to binding posts on the DC power entrance panel. See 3.11.3.5.

3.9.8.6 Grounding. Reference Electronics Command Drawing SC-B-61578.

3.9.8.6.1 DC circuits: The negative side of the nominal 28 volt DC power feed shall be grounded to the frame of the shelter and to the ground bus bar within the shelter.

3.9.8.6.2 AC circuits: All AC electrical outlets shall have two contacts for power, and a third contact for ground (green wire). All ground contacts in AC outlets shall be connected to the frame of the shelter, and to the ground lead (green wire) in the main AC power receptacle on the AC power entrance panel. Neither side of the two wire AC power feed to Government furnished and contractor furnished AC operated items shall be grounded. The wires at the free end of the 50 foot A.C. primary power cable shall be marked as follows: BLACK = "115 VAC HOT"; WHITE = "115 VAC RETURN" and GREEN = "FRAME GROUND". Connections at the shelter end of the power cable shall be as follows: BLACK = A; WHITE = B, and GREEN = C.

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3.9.8.6.3 Ground Bus: A ground bus shall be installed in the shelter to provide grounding points for radio and other electronic equipment. This ground bus shall be connected to a threaded ground stud on the DC power entrance panel. The following items installed in the shelter shall be connected to the ground bus by flexible ground straps of minimum practicable length:

a. Receiver-Transmitter RT-662/GRC, Amplifier RF AM-3349/GRC-106 and associated Mounting MT-3140/GRC-106 as an integrated assembly.

b. Modem, Radio Teletypewriter and associated Mounting MT-3140/GRC-106 as an integrated assembly.

c. Inverter.

d. Teletypewriter Reperforator-Transmitter TT-76B/GGC.

e. Teletypewriter TT-98/FG.

f. Multi-fuel heater.

g. Power Supply PP-4763/GRC.

h. Auxiliary inverter (not part of AN/GRC-142( )).

i. Auxiliary Teletypewriter TT-98/FG (not part of AN/GRC-142( )).

j. Auxiliary RT-662/GRC RF Preselector TN-510/GRC (not part of AN/GRC-142( )).

3.9.8.7 Wherever practicable, wiring and cabling shall be inclosed in wire mold, designed and installed so as to be free of sharp edges or points within the wire mold which could damage or puncture wire insulation. Where the use of wire mold is not practicable, wiring and cabling shall be held in place by an approved means to prevent damage to cables and wiring due to vibration, shock, or movement. Signal and power cable for teletypewriter equipment shall be provided with a positive means for preventing damage to these cables when teletypewriters are moved on their slide mounting bases from one extreme of travel to the other. The AN/GRC-106 whip antenna lead-in shall be supported from the ceiling of the shelter by a means which will withstand the RF voltage present on the lead-in wire, and allow removal of the AM-3349/GRC-106 without interference or the use of tools.

3.9.9 EMI Characteristics: The radio teletypewriter set shall be tested in accordance with 4.10.10 using the approved test plan of 4.3.3. In each of the tests required for compliance, the data must show that the first article sample is equal to

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or better than the like data for the procurement reference model or equal to or better than the requirements of MIL-STD-461. In the event that the first article sample must be compared to the procurement reference model, the contractor is responsible for the testing of the procurement reference model. The method for test of all models shall be as nearly identical as possible and all data shall be recorded, analyzed, presented in graph form where possible and included in the compliance report per 4.4. In addition, the contractor is required to perform additional tests for information purposes as outlined in 4.10.10.

3.9.10 Federal Standard -222 Compliance: The equipment shall meet the Tempest requirements of FED-STD-222 to the maximum extent possible utilizing tempest procedures and techniques similar to those incorporated in the procurement model. Reference to equipment under test, as cited in Standard FED-STD-222 shall be considered to apply to the AN/GRC-142( ) assemblage as a single unit. See 4.3.2 and 4.10.11.

3.9.11 System wiring and cabling for S-318/G configuration: System wiring and cabling in the S-318/G configuration of the AN/GRC-142( ) shall be provided in accordance with the drawings.

3.10 Physical and Mechanical Safety Measures.

3.10.1 Seat Belt and Shoulder Harness: The operator's chair, 3.26, shall be equipped with an approved type of seat belt and shoulder harness, suitably secured to the interior of the shelter.

3.10.2 Safety Flags: Red triangular flags with fastening devices shall be provided for the ends of the two whip antennas which may be mounted on the shelter.

3.10.3 Blunt End Fittings for Whip Antennas: Two each blunt end fittings shall be supplied for attachment to the tip ends of the two whip antennas which may be mounted on the shelter.

3.10.4 Fire Extinguisher and Bracket: An approved type fire extinguisher with appropriate bracket shall be installed in the shelter near the door.

3.10.5 Vehicle and Heater Exhaust Hoses: Flexible fireproof hoses, with suitable means for attachment, shall be provided to duct away the vehicle exhaust fumes and the shelter heater exhaust fumes. Each flexible hose shall be ten (10) feet long. Instruction plates shall be affixed to the outside of the shelter to show the method for attachment of each flexible hose. The vehicle exhaust extension hose shall be designed to attach to the M37 truck tail pipe collar plate as well as to the collar plate extension stub, provided on some models of the M37 truck.

3.10.6 Mechanical Safety: The equipment shall be constructed so that when assembled and set up for operation, accidental contact by personnel with moving parts such as gears, fans and belts shall not be possible. Doors, hinged panels and other contractor furnished parts shall be free of sharp edges which could cause injury to personnel during operation and maintenance.

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3.11 Shelter, Electrical Equipment S-318/G: Shelter, Electrical Equipment S-318/G shall be furnished by the Government. Performance of the shelter as a result of changes and/or additions shall not be degraded below that required of the shelter Specification MIL-S-55429.

3.11.1 Shelter Heating and Ventilating Facilities.

3.11.1.1 Ventilating Fan: A ventilation fan, 28v dc, IMC Magnetics Corporation, Westbury, Long Island, New York, Part Number BD-3020F-1 or equal shall be provided to exhaust the air in the shelter. The exhaust vent shall have a reusable filter. In order to reduce noise generated by the ventilation fan, the ventilation fan shall be mounted on vibration isolators and the blower plenum shall be undercoated. The fan shall be mounted on the front wall (cab end) of the shelter. A plenum shall be installed on the front interior wall of the shelter to provide a uniform "draw" of air through the shelter. Means shall be provided to allow ready access to the ventilation fan for maintenance. A rain proof hood, with RFI filter and a reusable, removeable dust filter, shall be installed on the outside of the shelter over the fan exhaust port.

3.11.1.2 Multi-fuel Heater: A multi-fuel heater, 28 vdc, type III in accordance with MIL-H-52204 shall be installed in the shelter together with associated fuel line, thermostat, fuel tank, fuel tank adapter, intake and exhaust ports and piping. The heater shall be mounted on the roadside step in the shelter, near the door. The thermostat shall be mounted on the curbside wall near the ceiling. The 5 gallon fuel tank shall be mounted on the outside rear wall of the shelter, to the roadside of the door.

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3.11.2 Major Equipment Mounting Facilities.

3.11.2.1 Main Equipment Mounting Rack: A main equipment mounting rack shall be assembled and installed in the shelter, across the front (cab end) wall. This rack shall be equipped with shelves and mounting facilities as follows:

a. Left hand section, top to bottom:

(1) Shelf for mounting MT-3140/GRC-106 with RT-662/GRC and AM-3349/GRC-106 attached thereto.

(2) Facilities for panel mounting of SA-1555/GRC-142.

(3) Shelf, with slides, for mounting TT-76( )/GGC.

b. Center section, top to bottom:

(1) Shelf for mounting MT-3140/GRC-106 with MD-522A/GRC and auxiliary RT-662/GRC attached thereto. (Auxiliary RT-662/GRC is not supplied as part of AN/GRC-142).

(2) Facilities for mounting SA-1554/GRC-142.

(3) Shelf, with slides, locks and mounting base for Mounting TT-98/FG.

(4) Facilities for mounting AC/DC power transfer switch and second LS-166A/U.

(5) Base and hardware for mounting PP-4763/GRC, per SM-D-602939 and SM-C-603830.

(6) Storage compartments for teletypewriter paper rolls and tapes, to the left of the PP-4763/GRC.

(7) Base for mounting main and auxiliary inverters to the right of the PP-4763/GRC. (Auxiliary inverter is not supplied as part of AN/GRC-142).

c. Right hand section, top to bottom:

(1) Facilities for mounting SB-3018/GRC-142, hinged on left side of panel face.

(2) Facilities for mounting blackout curtain.

(3) Shelf with guides and durable hand-tightened clamps for mounting J-2728/GRC-142 and auxiliary security device. (Auxiliary security device is not supplied as part of AN/GRC-142).

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(4) Facilities for mounting LS-166A/U and C-434/GRC.

(5) Shelf with guides and durable hand tightened clamps for mounting J-2728/GRC-142 and security device (Security device is not supplied as part of AN/GRC-142).

3.11.2.2 Roadside step: A metal storage compartment with hinged doors employing simple, rugged "turn" type latches shall be mounted on the roadside step. The multi-fuel heater shall be mounted on top of the storage compartment toward the rear of the shelter. Slides shall be mounted on the top of the storage compartment, forward of the heater, to enable the TT-76( )/GGC to be pulled into its operating position, and locked, or pushed back into its stored position, in the left hand section of the main equipment rack, and locked. See Electronics Command Drawing List SC-DL-603311.

3.11.2.3 Curbside step: A metal storage compartment with hinged doors employing simple, rugged "turn" type latches shall be mounted on the curbside step. A mounting base for the auxiliary TT-98/FG with slides, shall be secured to the top of the storage compartment. Means shall be provided to lock the auxiliary TT-98/FG mounting base in position at the extreme ends of its forward and aft travel. (The auxiliary TT-98/FG is not supplied as part of AN/GRC-142). Signal cables at the auxiliary TT-98/FG position shall be supported by a simple means which will prevent shock hazards and shorted terminals. See Electronics Command Drawing List SC-DL-603312.

3.11.3 Miscellaneous items mounted in and on Shelter S-318/G.

3.11.3.1 Fuel Container Bracket and Strap Assembly: Means shall be provided on the rear wall, roadside of the door, for rattle-proof mounting the fuel container, 5 gallon capacity, and its bracket. A footman loop, shall be mounted to the shelter wall, above the fuel container, for the web strap which is supplied with the bracket.

3.11.3.2 Box, Telephone: A metal box to hold Telephone, TA-312/PT shall be mounted on the curbside wall, near the main equipment rack. See Electronics Command Drawing List SC-DL-603313.

3.11.3.3 Front Whip Antenna Mounting Bracket: A bracket, with pivoted section to accommodate Mast Base AB-652/GR shall be mounted on the front wall of the shelter, just below the roof line, near the road side, to accommodate the AN/GRC-106 whip antenna assembly. See also 3.11.3.4.1 and Electronics Command Drawing List SC-DL-49678 - Mast Base AB-652/GR.

3.11.3.4 Rear Whip Antenna Mounting Bracket: A bracket, with pivoted section to accommodate Mast Base AB-652/GR shall be mounted on the rear wall of the shelter, just below the roof line, near the curb side, to accommodate the auxiliary RT-662/GRC receiver transmitter used in full duplex operation (auxiliary RT-662/GRC is not supplied as part of AN/GRC-142). See also 3.11.3.4.1.

3.11.3.4.1 The whip antenna mounting brackets, 3.11.3.3 and 3.11.3.4 shall be adjustable to two positions. Position Number 1 shall support the whip antenna at an angle 10 degrees from vertical so that the antenna will tilt away from the shelter. Position Number 2 shall support the whip antenna at an angle 90 degrees from vertical, for installing or servicing whip antenna masts. The mounting brackets shall have stops and locking pins for each of the two positions mentioned above.

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3.11.3.5 DC Power Entrance Panel: A DC power entrance panel shall be installed on the front (cab end) wall of the shelter, just below the roof line, near the curb side. This DC power entrance panel shall provide a receptacle for connection of 28 volt DC 100 ampere service, and five pairs of binding posts for remote field wire lines. The DC power entrance panel shall include a clamp for stowing the 10 pin audio connector of Cable CX-1852/U used in remote CW service, a ground stud, lightning arrestors, RFI filters and terminals for connections to shelter interior wiring, including, Cable, Special Purpose CX-1852/U which connects E11 and E12 to the "VOICE-KEY" receptacle of Control Panel SA-1554/GRC-142 during remote CW service. See Electronics Command Drawing List SC-DL-603309.

3.11.3.6 AC Power Entrance Panel: An AC power entrance panel shall be installed on the rear wall of the shelter, curb side of the door just above the level of the curb side step. This AC power entrance panel shall provide a receptacle for connection of 115 volt 50/60 HZ single phase AC 60 ampere service. Two circuit breakers and terminations for AC branch circuits in the shelter. See Electronics Command Drawing List SC-DL-603310.

3.11.3.7 Fuel Container Bracket: A fuel container bracket shall be mounted on the rear wall, using the means provided per paragraph 3.11.3.1.

3.11.3.8 Ducting, piping and covers for heater intake and exhaust ports: The heater exhaust pipe shall be extended through the roadside wall, with appropriate framing and fireproof gasketing. A cover, captivated by a chain, and equipped with means for rattleproof storage when not connected to the exhaust pipe, shall be provided. The heater air intake port shall be ducted through the rear wall of the shelter, with appropriate framing. A cover, captivated by a chain and equipped with means for rattleproof storage when not secured over the air intake port duct, shall be provided. The heater exhaust pipe shall be equipped with means for attaching the flexible extension hose, paragraph 3.10.5.

3.11.3.9 Rubber covers for remote line binding posts: The ten binding posts on the DC power entrance panel shall be equipped with rubber covers. Red rubber covers shall be used on the positive posts. Black rubber covers shall be used on the negative posts.

3.11.3.10 Physical security requirements.

3.11.3.10.1 Facilities as described below shall be provided for locking the inner and outer door of the shelter:

a. Retain the handles on the inside and outside of the shelter outer door.

b. Provide means for locking the outside handle of the shelter outer door, using a three position dial type combination padlock which meets the requirements of MIL-P-17257 . The dial combination padlock will not be furnished as part of the AN/GRC-142.

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c. Retain the handle on the inside of the shelter inner door.

d. Delete the outside handle on the shelter inner door and seal against rain and intrusion.

e. Provide a means for locking the outer and inner doors from the inside of the shelter.

f. Door hinges on inner and outer doors shall be modified to insure that mounting hardware and hinge pins cannot be removed without visible damage to these parts.

3.11.3.10.2 Night latch: A chain type night latch, #440, as supplied by Baldwin Hardware Mfg. Co., or equal, shall be provided on the interior of the shelter for the shelter outer door.

3.11.3.10.3 Door viewer: The shelter inner door shall be provided with an angle optical lens #D-50 as supplied by Call Boy Co., or equal.

3.11.3.10.4 Mail slot and message tray: A mail slot shall be provided on the rear wall, roadside of the door. The mail slot shall be equipped with a waterproof cover, with hinges and hand tightened fasteners. The opening shall be provided with a means for maintaining the RFI integrity of the shelter. The slot shall not permit viewing of the interior of the shelter from the outside. A message tray shall be installed in the shelter under the mail slot.

3.11.3.10.5 Blackout curtain: A blackout curtain shall be provided to cover the right hand section of the main equipment rack, 3.11.2.1. A tie back shall be provided for the curtain. The curtain shall be hung from the rod provided below the Power Distribution Panel, SB-3018/GRC-142, and shall extend to the bottom of the lower shelf which supports the security device. The curtain and its tieback shall be of fireproof material.

3.11.3.10.6 Door chime: A rust and corrosion proof mechanically operated door chime shall be mounted on the rear wall to the roadside of the door. Means shall be provided to prevent entrance of water through the door chime mechanism.

3.11.3.10.7 Door interlock switch: The shelter outer door shall be equipped with an electrical interlock switch, so that when the door is opened, the main electrical lighting circuit is broken before the controlled light is visible through the opened door. The interlock switch shall be provided with a manual over ride feature which automatically resets upon closing the door.

3.11.3.11 R.F. Preselector (TN-510/GRC) mounting facilities: Facilities shall be provided in the shelter for installing and operating an RF Preselector in conjunction with the auxiliary RT-662/GRC, for full duplex operation. Mounting provisions for the Preselector shall be sturdy enough to allow the Preselector to withstand the rail hump test, road test and air drop test without damage and/or dislocation. The Preselector shall be mounted near the ceiling, at the rear curbside corner of the shelter. A model of the Preselector will be Government furnished to the contractor. The Preselector and auxiliary RT-662/GRC are not part of the AN/GRC-142. Mounting facilities for the Preselector shall afford ease of installation and removal, without the use of tools.

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3.11.3.12 Hanger clamp for steel helmet: A suitable clamp shall be mounted at a convenient location inside the shelter to hold a soldier's steel helmet. The clamp shall be properly identified as to function.

3.11.3.13 Mounting provisions for Bag CW-206/GR: Web straps shall be secured to the vertical member of the interior curb side step, to hold Bag CW-206/GR (supplied as part of AN/GRC-106). Means for securing the seb straps to the step shall be sufficiently sturdy to hold the CW-206/GR in place during and after rail hump test and parachute drop test.

3.11.3.14 Hanger for H-33/PT handset: A hanger for the H-33/PT handset shall be provided on a vertical member of the main equipment rack, near the control panel, and out of the way of all teletypewriters, to prevent the handset cord from interfering with operation of the teletypewriters. The handset hanger shall be suitably marked by an identifying label.

3.11.3.15 Hanger for M-29/U microphone: A hanger for the M-29/U microphone shall be provided on a vertical member of the main equipment rack, near the control panel, and out of the way of all teletypewriters, to prevent the microphone cord from interfering with operation of the teletypewriters. The microphone hanger shall be suitably marked by an identifying label.

3.11.3.16 Exhaust duct for AN/GRC-106 power amplifier: Means shall be provided for ducting the exhaust air from R.F.Amplifier AM-3349/GRC-106 to the main plenum (3.11.1.1) in the area of the ventilating fan. Ducting for the AM-3349/GRC-106 shall insure a tight fit to the exhaust port on the amplifier, and shall permit removal of the AM-3349/GRC-106, in its case, without the use of tools.

3.11.3.17 Writing surface/shelf extender: A combination writing surface/shelf extender shall be provided. This item shall be mounted above the storage compartment on the road side of the shelter (3.11.2.2). A clamp with hand operated release, shall be employed to hold the item against the wall when not in use. In use, as a writing surface, the item shall swing down and rest on the TT-76( )/GGC slides. In use as a shelf extender, the item shall mount on pins located at the front edge of shelves in the main equipment rack (3.11.2.1). The leg braces shall be connected by a horizontal cross member which is located so as not to interfere with security devices, when installed (3.11.2.1c).

3.11.3.18 Floor attachment for operator's chair: Means shall be provided for attachment of the operator's chair (3.26) to the shelter floor. It shall be possible to lock the chair to the floor without the use of tools. When locked to the floor, the chair shall be capable of passing the rail hump test without becoming unlocked or detached from the floor. The means for attachment of the operator's chair to the floor shall incorporate a quick-release feature, to allow the chair to be unlocked from the floor without the use of tools. Means shall be provided for adjustment of the tension of the locking mechanism, without the use of tools (see also 3.26).

3.11.3.19 Adjustment tool for Mounting MT-3140/GRC-106: Each shelter shall be equipped with a tool for adjusting cross bars on Mounting MT-3140/GRC-106, with the equipment installed on the mounting, and all other nearby items in place. This tool shall serve for adjusting cross bars on the two MT-3140/GRC-106 mounting bases in the shelter.

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3.11.3.20 Mounting and tie downs for shade tarpaulin: Hinged supports shall be provided on the curb side and road side of the shelter, near the roof, for attachment of the shade tarpaulin, (3.37). Eylets (four each) shall be installed on the road side of the shelter to accommodate the four tie-down ropes used to keep the shade tarpaulin taut. These eylets shall be located in a line just above the road side step.

3.11.3.21 Mounting clamps for ground rods: Means shall be provided on the outside of the shelter, under the road side step, for stowing two ground rods, (3.34) Clamps, operable without the use of tools shall be used to hold the ground rods in place.

3.11.3.22 Feed-through receptacles and fittings for R.F.cables: An R.F. feed-through receptacle, type N to type N shall be mounted in the road side wall of the shelter, above the heater exhaust port, to provide interconnection of the AN/GRC-106 to a doublet antenna. An R.F. feed-through receptacle, BNC type inside the shelter and type N outside the shelter, shall be mounted in the rear wall of the shelter, at the rear whip antenna bracket location, to provide interconnection of the R.F. Preselector to the antenna used for receiving, in full duplex operation. R.F. Preselector and auxiliary antennas are not part of AN/GRC-142. A feed-through insulator shall be mounted in the front wall of the shelter, at the front whip antenna location to provide interconnection of the AN/GRC-106 to a whip antenna, using the whip antenna lead-in cable provided as part of the AN/GRC-106. All feed-through provisions shall be light tight and water tight. Feed-through receptacles for coaxial cables shall not protrude beyond the exterior wall surface of the shelter. Captivated covers shall be provided on the outside of the shelter for the two coaxial cable feed-through receptacles. Means shall be provided to prevent these covers from rattling when they are removed from the receptacles.

3.11.3.23 Mounting provisions for cording diagram: Provisions shall be made on the interior curb side wall, near the rear of the shelter, for mounting the cording diagram (3.38).

3.11.3.24 Weapons holder: A clamp to retain the stock of an M-14 rifle, shall be mounted inside the shelter, near the floor, and to the curbside of the door.

3.11.3.25 Mounting provisions for ME-165/G Radio Power Meter: Provisions shall be made on the interior road side wall, near the ceiling, and toward the rear of the shelter, for mounting Radio Power Meter ME-165/G.

3.11.3.26 Message holder: A message holder shall be provided, and installed on the interior road side wall, to the right of the Radio Power Meter ME-165/G, and above the clock.

3.11.3.27 Mounting provisions for sledge hammer: Means shall be provided on the vertical member of the interior curb side step, to hold the sledge hammer, (3.35). The means provided shall be sturdy enough to hold the sledge hammer in place during and after the rail hump test and parachute drop test.

3.11.3.28 Special markings and labels: The special markings and labels as listed below, shall be provided and mounted in and on the shelter, to aid operator and maintenance personnel:

a. Placard with instructions for adjusting long and short cross bars on Mounting MT-3140/GRC-106.

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b. Placard with instructions for connecting CX-1852/U to Control Panel during remote CW keying. See 3.11.3.5.

c. Placard with information regarding type and maintenance requirements for fire extinguisher.

d. Durable marking bands for nomenclatured cables and wires, secured so as not to become loose from the cables and wires during operation and maintenance, under all service conditions.

e. The exterior of the shelter shall include markings to show center of gravity, (3 planes), points for lifting, hoisting, tie-down and fork lift, as well as gross weight of the shelter, with all equipment installed and/or stowed. This shall include markings for C.G. when configured as an AN/GRC-142/122.

f. Placard to identify location of M-29/U microphone bracket.

g. Placard to identify location of H-33/PT handset bracket.

h. Placard to identify location of hanger clamp for steel helmet.

i. Placards, mounted in appropriate locations, to show stowage space and mounting provisions for loose parts and components, and technical literature.

j. Placard to illustrate use of writing surface/shelf extender.

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3.12 Radio Set AN/GRC-106: Receiver-Transmitter RT-662/GRC and R.F.Amplifier AM-3349/GRC-106, major components of the AN/GRC-106, shall serve as the basic radio transmitting and receiving facilities in the AN/GRC-142( ) system. The RT-662/GRC and AM-3349 shall be installed and interconnected into the AN/GRC-142( ) system in accordance with applicable paragraphs of this specification. The RT-662/GRC and AM-3349/GRC-106 shall operate in the system in accordance with paragraphs 3.6, 3.7, 3.8 and 3.9. Mounting MT-3140/GRC-106, audio accessories, antenna equipment and miscellaneous items which are supplied as part of Radio Set AN/GRC-106, shall be incorporated into the AN/GRC-142( ) system in accordance with applicable paragraphs of this specification. Radio Set AN/GRC-106 shall be furnished by the Government.

3.13 Radio Teletypewriter Modem MD-522A/GRC: The MD-522A/GRC, a Government furnished item, shall serve as a means for processing voice and teletypewriter signals transmitted and received through Radio Set AN/GRC-106. This modem shall also serve as a means for processing voice and teletypewriter signals received by an auxiliary RT-662/GRC, in the full duplex mode of operation. The auxiliary RT-662/GRC is not supplied as part of the AN/GRC-142( ). The MD-522A/GRC shall be installed and interconnected into the AN/GRC-142( ) system in accordance with applicable paragraphs of this specification. The MD-522A/GRC shall be secured to its mounting base, MT-3140/GRC-106, normally by means of a pair of short cross bars, 3.32. When the auxiliary RT-662/GRC is employed for full duplex operation, this RT unit shall be mounted atop the MD-522A/GRC, and the stacked assembly shall be secured to the MT-3140/GRC-106 by means of a pair of long cross bars, not supplied as part of AN/GRC-142( ). The MD-522A/GRC shall operate in the system in accordance with paragraphs 3.6, 3.7, 3.8 and 3.9.

3.14 Teletypewriter TT-98/FG: The TT-98/FG, a Government furnished item, shall be installed and interconnected into the AN/GRC-142( ) system, in accordance with applicable paragraphs of this specification. The TT-98/FG shall operate in the system in accordance with the requirements of paragraphs 3.6, 3.7, 3.8 and 3.9.

3.15 Teletypewriter Reperforator-Transmitter TT-76B/GGC (or TT-76C/GGC): The TT-76B/GGC (or TT-76C/GGC), a Government furnished item, shall be installed and interconnected into the AN/GRC-142( ) system, in accordance with applicable paragraphs of this specification. The TT-76B/GGC (or TT-76C/GGC) shall operate in the system in accordance with the requirements of paragraphs 3.6, 3.7, 3.8 and 3.9.

3.15.1 Teletypewriter Reperforator-Transmitter TT-76/GGC and TT-76A/GGC: The TT-76/GGC and TT-76A/GGC, although similar in outward appearance to the TT-76B/GGC and TT-76C/GGC, and supplied under the same Federal Stock Number, 5815-553-6061, are NOT usable in the AN/GRC-142( ), since the TT-76/GGC and TT-76A/GGC will not accomodate Low Level Signalling Device TT-523A/GGC, paragraph 3.23.

3.16 Inverter, Motor Generator: The motor generator inverter, a Government furnished item, shall be installed and interconnected into the AN/GRC-142( ) system, in accordance with applicable paragraphs of this specification. This item shall serve to convert 28 volts DC power into 115 volt single phase AC power for operation of the TT-98/FG and TT-76B/GGC (or TT-76C/GGC) teletypewriters.

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3.17 Switch Box Assembly SA-1554/GRC-142: The SA-1554/GRC-142 (Control Panel) shall provide the following functions in the AN/GRC-142( ) system: See Electronics Command Drawing List SC-DL-603314.

a. D.C. teletypewriter circuit loop current adjustments (send and receive).

b. Local-Remote switching.

c. Lockout override switching.

d. "VOICE-KEY" receptacle for local audio accessories or for remote CW cable CX-1852/U.

e. Field telephone call indicator.

f. Audio and control terminations for local audio accessories as well as for remote control (AN/GRA-6) equipment, C-434/GRC, per latest revision to Drawing SM-D-602945.

g. Teletypewriter loop terminations for connection to two each J-2728/GRC-142 Interconnecting Box (Dummy Box).

h. Audio and control terminations for connection to Modem MD-522A/GRC.

i. Teletypewriter loop terminations for connection to Modem MD-522A/GRC.

j. Terminations for connection to D.C. Power entrance panel.

k. Terminations for connection to TA-312/PT telephone, mounted in the shelter.

l. Terminations for connection to SB-3018/GRC-142 (Power Panel) and two each J-2728/GRC interlocks. The SA-1554/GRC-142 shall be installed and interconnected into the AN/GRC-142( ) system in accordance with applicable paragraphs of this specification. The SA-1554/GRC-142 shall provide mounting facilities for ease in mounting and removal from the equipment rack without disturbing other equipments. It shall be located in a manner not to interfere with the operation of the teletypewriter equipment.

3.18 Panel, Power Distribution, SB-3018/GRC-142: The SB-3018/GRC-142 (Power Panel) shall provide the following functions in the AN/GRC-142( ) system: See Electronics Command Drawing List SC-DL-603315.

a. Current overload protection (circuit breaker) in the primary 28 volt DC feed circuit from the vehicle electrical system.

b. Indication of primary DC voltage (0-50 volt D.C. voltmeter).

c. Indication of primary DC voltage (Indicator lamp, 28 VDC with red lens).

d. Current overload protect (fuses) and 28 VDC branch circuit on-off switching for:

(1) Shelter lights

(2) Main ventilating fan, with "full speed-off-half speed" switch and associated dropping resistor.

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(3) Lockout relay

(4) 28 VDC convenience outlet

(5) Main inverter-motor generator

(6) Auxiliary inverter-motor generator (not suppled as part of AN/GRC-142( ) system).

e. Terminations for main and branch 28 volt DC power circuits.

The SB-3018/GRC-142 shall be installed and interconnected into the AN/GRC-142( ) system in accordance with applicable paragraphs of this specification. The SB-3018/GRC-142 shall include a hinge along the right hand side of the front panel. The left hand and bottom edges of the hinged front panel shall be cushioned to prevent injury to personnel during maintenance operations which require the panel to be swung out on its hinge.

3.19 Switch Box, SA-1555/GRC-142: The SA-1555/GRC-142 (Red-Black Switch Box) shall provide the following functions in the AN/GRC-142( ) system: See Electronics Command Drawing List SC-DL-603316.

a. "Red-Black"switching for one way reversible and duplex sending teletypewriter circuits.

b. "Red-Black" switching for duplex receiving and "pony" teletypewriter circuits.

c. Tape punch switching between one way reversible/duplex sending teletypewriter circuits and duplex receiving teletypewriter circuits.

d. Teletypewriter loop terminations for connection to:

(1) TT-76B/GGC (or TT-76C/GGC) Transmitter Distributor.

(2) TT-98/FG Keyboard.

(3) Auxiliary TT-98/FG Keyboard

(4) Auxiliary TT-98/FG Selector Magnets

(5) TT-98/FG Selector Magnets

(6) TT-76B/GGC (or TT-76C/GGC) Selector Magnets

(7) J-2728/GRC-142 (OWR/DX SEND) Interconnecting Box send circuit

(8) J-2728/GRC-142 (DX RCV/PONY) Interconnecting Box send circuit

(9) J-2728/GRC-142 (OWR/DX SEND) Interconnecting Box receive circuit

(10) J-2728/GRC-142 (DX RCV/PONY) Interconnecting Box receive circuit

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The SA-1555/GRC-142 shall be installed and interconnected into the AN/GRC-142( ) system in accordance with applicable paragraphs of this specification. The SA-1555/GRC-142 shall provide mounting facilities for ease in mounting and removal from the equipment rack without disturbing other equipments. It shall be located in a manner not to interfere with operation of the teletypewriter equipment.

3.20 Interconnecting Box J-2728/GRC-142: Two (2) each J-2728/GRC-142 (Dummy Box) shall be used in the AN/GRC-142( ) system. See Electronics Command Drawing List SC-DL-603317.

3.20.1 One (1) each J-2728/GRC-142 shall provide the following functions in the AN/GRC-142( ) system:

a. Teletypewriter loop terminations for connection to the SA-1555/GRC-142 OWR/DX SEND transmit circuits.

b. Teletypewriter loop terminations for connection to the SA-1555/GRC-142 OWR/DX SEND receive circuits.

c. Teletypewriter loop terminations for connection to SA-1554/GRC-142 OWR/DX SEND circuits.

d. The J-2728/GRC-142 when bolted in place on the lower shelf in the right hand section of the main equipment rack acts to close the associated interlock switch. Placement of the J-2728/GRC-142 on the lower shelf prevents installation of the OWR/DX SEND teletypewriter security device.

e. Removal of the J-2728/GRC-142 from the lower shelf acts to open the associated interlock switch and cause radio telephone and wire line voice circuits to be "locked out".

3.20.2 The second J-2728/GRC-142 shall provide the following functions in the AN/GRC-142( ) system:

a. Teletypewriter loop terminations for connection to the SA-1555/GRC-142 DX RCV/PONY transmit circuits.

b. Teletypewriter loop terminations for connection to the SA-1555/GRC-142 DX RCV/PONY receive circuits.

c. Teletypewriter loop terminations for connection to SA-1554/GRC-142 DX RCV/PONY circuits.

d. The J-2728/GRC-142, when bolted in place on the upper shelf in the right hand section of the main equipment rack, acts to close the associated interlock switch. Placement of the J-2728/GRC-142 on the upper shelf prevents installation of the DX RCV/PONY teletypewriter security device.

e. Removal of the J-2728/GRC-142 from the upper shelf acts to open the associated interlock switch and cause radio telephone and wire line voice circuits to be "locked out". Each J-2728/GRC shall be installed and interconnected into the AN/GRC-142( ) system in accordance with applicable paragraphs of this specification.

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3.21 Telephone, TA-312/PT: Two each TA-312/PT telephones shall be used in the AN/GRC-142( ) system. These items shall be Government furnished. One telephone shall be stowed in the shelter, for use during remote operations, per paragraph 3.6.2. The other telephone shall be installed in the box type holder, paragraph 3.11.3.2, and interconnected into the AN/GRC-142( ) system, in accordance with applicable paragraphs of this specification.

3.22 Radio Power Meter ME-165( )/G: The ME-165( )/G, a contractor-furnished item, shall be installed and interconnected into the AN/GRC-142( ) system, in accordance with applicable paragraphs of this specification. This meter shall serve as a dummy R.F. load for Radio Set AN/GRC-106, and as an indicator of VSWR on the coaxial cable used to connect the AN/GRC-106 to a doublet antenna. Reference Military Specification MIL-M-55234 and Electronics Command Drawing SC-DL-349717.

3.23 Low Level Signalling Device TT-523A/GGC: The TT-523A/GGC and associated 5,600 ohm resistor, Government furnished items, shall be stowed in the shelter for use during secure radio teletypewriter operation. These two items are intended for installation on Teletypewriter Reperforator-Transmitter TT-76B/GGC (or TT-76C/GGC). Reference Electronics Command Drawing SM-D-603491.

3.24 Power Distribution Terminal Board: The power distribution terminal board, with housing and cover, shall be installed on the top shelf of the center section of the main equipment rack, behind the MT-3140/GRC-106. This terminal board shall serve as a distribution point for primary power feed to the 28 volt DC operated items as listed below:

a. Receiver-Transmitter RT-662/GRC.

b. R.F. Amplifier AM-3349/GRC-106.

c. Modem MD-522A/GRC.

d. Auxiliary Receiver-Transmitter RT-662/GRC. (Power cable is supplied as part of AN/GRC-142( ) system. Auxiliary RT-662/GRC is not supplied as part of AN/GRC-142( ) system). DC power to the terminal board shall be drawn from appropriate terminals in the Power Distribution Panel SB-3018/GRC-142. The power distribution terminal board shall be installed and interconnected into the AN/GRC-142 ( ) system in accordance with applicable paragraphs of this specification.

3.25 Power Supply PP-4763/GRC: The PP-4763/GRC, a Government furnished item, shall be installed and interconnected into the AN/GRC-142( ) system, in accordance with applicable paragraphs of this specification. This power supply shall serve to convert 115 volts 50/60 Hz single phase primary power into 28 volts DC for operation of certain components in the AN/GRC-142( ) system, when the system is operating from AC primary power.

3.26 Operator's chair: A padded swivel chair with casters (steno type) shall be provided. It shall be installed in front of the teletypewriter TT-98/FG and shall be provided with a mechanism so that when made ready for transport, it can be securely clamped to the shelter floor. This mechanism shall be a quick release type device, and the portion installed in the floor shall store to a position flush with the floor. The operator's chair shall be equipped with an approved type of seat belt and exception to model.

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3.27 Clock: A clock, Aircraft type, Longines Wittnauer type A-11, or equal, shall be provided. This clock shall be mounted on the roadside wall of the shelter, below the message holder.

3.28 Control Group AN/GRA-6: The AN/GRA-6, a Government furnished item, shall be used in the AN/GRC-142( ) system. The local control, C-434/GRC and speaker, LS-166A/U shall be installed and interconnected into the AN/GRC-142( ) system in accordance with applicable paragraphs of this specification. Other components of the AN/GRA-6 shall be stowed for use during remote operations (per paragraph 3.6.2). The LS-166A/U shall be used as a monitor speaker for the auxiliary RT-662/GRC used in full duplex operation. The auxiliary RT-662/GRC is not supplied as part of the AN/GRC-142( ).

3.29 Remote Control, C-7279/GRC-142: The C-7279/GRC-142 (remote Box) shall provide the following functions during remote operation of the AN/GRC-142( ) system: See Electronics Command Drawing List SC-DL-603308.

a. Send-receive switching for radio teletypewriter operation.

b. Field wire terminations for OWR/DX SEND remote teletypewriter D.C. signal circuit.

c. Field wire terminations for DX RCV/PONY remote teletypewriter D.C. signal circuit.

d. Teletypewriter loop terminations for connection to remote teletypewriters.

e. Audio and control termination for audio accessory at remote site.

f. Audio and control termination for connection to Control, C-433/GRC (remote control unit of AN/GRA-6).

In addition, the captivated protective cover for the audio receptacle on the C-7279/GRC-142 shall be painted black, corresponding to the color of the box. The C-7279/GRC-142 shall be stowed in the shelter for use during remote operation.

3.30 Fuel tank, 5 gallon capacity: The 5 gallon fuel tank, a Government furnished item, shall be mounted and secured in the bracket on the outside of the shelter, in accordance with applicable paragraphs of this specification. This fuel tank shall be used to store fuel for the multi-fuel heater (paragraph 3.11.1.2).

3.31 Fuel line adapter assembly: The fuel line adapter assembly, a Government furnished item, shall be stowed for use during cold weather operation as a means for feeding fuel from the fuel tank (3.30) to the heater fuel intake fitting, mounted on the exterior of the rear wall of the shelter.

3.32 Crossbar Assembly Short: Two each crossbar Assembly, Short, per Drawing List SC-DL-502745 shall be provided to hold Modem MD-522A/GRC in place on Mounting MT-3140/GRC-106, in the center section of the main equipment rack. Two each Crossbar Assembly, Long, are used to hold an auxiliary RT-662/GRC and Modem MD-522A/GRC in a stacked configuration on MT-3140/GRC-106 when operating

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in full duplex. The long crossbars and the auxiliary RT-662/GRC are not provided as part of the AN/GRC-142( ).

3.33 Rope clamp for MS-116A mast section: Three (3) each rope clamps, per Drawing SC-B-19491, shall be provided, and stowed when not in use. One clamp shall be used to secure a tie down rope (provided as part of AN/GRC-106) to the whip antenna (provided as part of AN/GRC-106). The other two clamps are spare items.

3.34 Ground Rod: Two (2) each ground rod, per Drawing SM-D-14158 shall be provided, and stowed when not in use, in brackets and fixtures provided on the outside of the shelter, per 3.11.3.21.

3.35 Sledge Hammer, 5 pound: A 5 pound sledge hammer, per Drawing SM-C-603093 shall be provided and stowed when not in use in brackets provided inside the shelter, per 3.11.3.27.

3.36 Mounting MT-3140/GRC-106: The MT-3140/GRC-106, a Government furnished item, shall be mounted on the top shelf of the center section of the main equipment rack. This mounting shall be used to secure Modem MD-522A/GRC in place, when clamped by the associated short cross bars, 3.32.

3.37 Tarpaulin, Shade.

3.37.1 One each shade tarpaulin, per Drawing SM-D-603394, complete with four tie down ropes, shall be provided for use with the S-318/G configuration of the AN/GRC-142( ).

3.38 Cording Diagram: A cording diagram, made of or protected by durable material shall be mounted on the interior wall on the curbside of the shelter. This cording diagram shall include connections and cabling information for the Power Supply PP-4763/GRC, auxiliary RT-662/GRC, R.F.Preselector, AC and DC exterior power cables, and all interior signal, power, control and RF cabling used in the AN/GRC-142( ) system, including cables used to provide AC and DC power to the security devices. Cables used in full duplex service, but not supplied as part of the AN/GRC-142 shall be shown in dotted lines.

3.39 Hand tools: The following tools shall be provided: Allen wrenches, 8 each assorted, screwdrivers, one 4" blade and one 6" blade, battery operated 4 cell hand lantern, heavy duty slip joint pliers, and small adjustable wrench (6" crescent or equal).

3.40 Parts, materials, and processes; general: In addition to the requirements of this specification, the requirements of Specification MIL-P-11268 including the selection requirements therein, shall apply and Military Specification MIL-C-26482 connectors electrical, circular miniature quick disconnect.

3.40.1 Cable assemblies: The length of electrical cable assemblies shall conform to Military Standard MS-75027. Power and audio frequency (af) cable and cord assemblies used for voltages up to and including 600 volts alternating current (ac) or corresponding dc shall conform to all requirements of Specification MIL-C-3885. Cable assemblies used for over 600 volts ac or equivalent dc shall be subject to approval by the Contracting Officer's Technical Representative. See Military Specification MIL-C-3432.

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Where applicable, high voltage cable shall conform to Specification MIL-C-3702. Radio frequency (rf) coaxial cables shall be in accordance with Specification MIL-C-17.

3.40.2 Finish, protective: The equipment shall be finished in accordance with Specification MIL-F-14072. This inclues finish of hardware, such as handles, screws, etc., and necessary touch-up after mounting.

3.40.2.1 The final paint film on Type I exterior surfaces shall be green color (olive drab) semigloss enamel matching color chip X24087 of Federal Standard 595.

3.40.2.2 Interior of shelter shall be painted light green #24533. (All color numbers refer to Federal Standard 595).

3.40.2.3 The inside of doors which open outward shall be painted the same color as the outside of the enclosure (Semi-gloss #24087).

3.40.2.4 Floor shall be painted lusterless, ocean gray #36118.

3.40.2.5 Equipment racks, storage compartments and components within the shelter except GFP shall be painted semi-gloss light gray enamel #26250.

3.40.3 Marking.

3.40.3.1 Conformance: Marking shall conform to Specification MIL-M-13231. Front panel marking shall be group I as described in that specification.

3.40.3.2 Visibility of marking: Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.40.3.3 Serial numbers: The following items shall have corresponding serial numbers:

a. Complete AN/GRC-142( ) shelter, fully equipped.

b. Switch Box Assembly SA-1554/GRC-142.

c. Panel, Power Distribution, SB-3018/GRC-142.

d. Switch Box, SA-1555/GRC-142.

e. Interconnecting Box J-2728/GRC-142.

f. Remote Control C-7279/GRC-142.

g. Radio Power Meter ME-165/G.

3.40.4 Relays: Where possible, all control relays used in the equipment shall be of the plug-in type, and shall be easily accessible for replacement.

3.40.5 Rotating knob end stops: All control knobs on contractor-furnished items shall withstand repeated force in each direction of 72 inch pounds without permanent deformation to the shaft and stops.

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3.40.6 Plastic materials and parts: Where not machined, plastic materials and parts shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable, in accordance with good manufacturing practice for the intended application.

3.40.7 Cleaning.

3.40.7.1 Parts: After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.40.7.2 Units: After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.40.8 Riveting: Thickness of countersunk metal which accepts the heads of flush rivets shall be not less than the height of the rivet heads. After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.40.9 Securing of parts.

3.40.9.1 Mounting components: Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall be such as to retain securely when subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent separation of parts will not occur when the items in which they are used are subjected to specified service conditions. Any permanent deformation resulting from the tests shall not cause degradation of performance below full specification requirements. Parts shall be secured in such manner that failure of a single rivet or screw will not free the part completely.

3.40.9.2 Brittle material: Brittle castings, or parts made of ceramic or other brittle material, shall not be over-constrained by the means used to secure them. Not more than three points of contact shall be used between the brittle part and the mounting surface, unless the mating faces are accurately machined. Mounting washers of suitable plastic, rubber, or soft copper having slight compressibility shall be provided to prevent breakage or cracking of the parts. Lead washers shall not be used. Threaded holes in ceramic material shall not be employed for assembly or mounting of parts except where voltage breakdown problems exist.

3.40.10 Brazing: Brazing of steels, copper, copper alloys, and nickel alloys shall conform to Specification MIL-B-7883.

3.40.11 Soldering.

3.40.11.1 Solder: Solder shall conform to Specification QQ-S-571. For soldering electrical connections, composition Sn60, type AR or S, shall be used for general purposes; and composition Sn62 or Sn63, type AR or S, for special purposes. When type S solder is used for soldering electrical connections, the flux shall conform to Specification MIL-F-14256.

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3.40.11.2 Acid or acid salts: No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.40.11.3 Process: There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristle, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.40.12 Tropicalization of material: Material shall be treated as follows:

3.40.12.1 Treating materials: Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.40.12.2 Toxicity: Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment or during use of the finished items when used for the purpose intended.

3.40.12.3 Flexibility: Treatment shall not affect the flexibility of treated materials to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.40.12.4 Statement of treatment: The contractor shall submit for approval, to the Contracting Officer, a statement describing in detail the materials to be treated and the treating material and processes that he proposes to use. (See 6.1(i).

3.40.13 Welding: Wherever practicable, welded joints shall conform to Standard MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858. Spot or other intermittent welds in aluminum or magnesium shall be limited to lightly stressed, non-critical structures. When spot or other intermittent welds are used to hold a part, the number of welds shall be at least two.

3.40.13.1 Cleaning prior to welding: Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

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3.40.13.2 Process: Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum, magnesium, or stainless steel.

3.40.13.3 Cleaning after welding: Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with Specification MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed.

3.40.14 Wiring and cabling: Wiring and cabling shall be neat and sturdy.

3.40.14.1 Cabling: Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance. The cabling of wires shall be effected by tape or twine conforming to Specification MIL-T-713. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit use of twisted pairs.

3.40.14.2 Slack: Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

a. To prevent undue stress on cable forms, wires, and connections, including connections to resilliently supported parts.

b. To enable parts to be removed and replaced during servicing without disconnecting other parts.

c. To provide for at least two replacements of the part to which the wire or cable is connected, except for certain rf leads which must be as short as possible for electrical reasons.

d. To ensure freedom of motion of lugs or terminals normally intended to have some degree of movement (for example, floating contacts on electron tube sockets).

e. To facilitate field repair of broken or cut wires.

f. To prevent chafing or breaking of individual wires due to repeated flexing of hinged parts. Reference Military Standard 33540.

3.40.14.3 Protection: Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material. Where wires run through holes in metal partitions, shields or similar items less than 1/8 inch in thickness, the wires shall be protected by suitable grommets or bushings. Where wires run through holes in metal 1/8

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inch in thickness or greater, the wires shall be protected by grommets or bushings or shall have the edges of the holes rounded to a minimum radius of 1/16 inch.

3.40.14.4 Support: Wire and cable shall be properly supported and secured, to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (I) during and after subjection of the equipment to specified service conditions, or (II) after service or repair of the equipment in a normal manner. (NOTE: When shielding on wire or cable is unprotected by an outer insulation, adequate support is necessary to prevent the shielding from coming in contact with exposed terminals or conductors). Twine or tape shall not be used for securing wire and cable.

3.40.14.5 Clearance: Clearance between wires or cables, and parts such as electron tubes, resistors, and converters, shall be provided to avoid deterioration of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.40.14.6 Splicing: Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.40.14.7 Insulating sleeving. Insulating sleeving shall be used only in accordance with Specification MIL-P-11268.

3.40.14.8 Inductive and capacitive effects: Wires and cables shall be so located that inductive and capacitive effects, unless used as a design feature, will be the minimum practicable.

3.40.14.9 Connections: Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver-plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and a non-metallic material. Welded electrical connections shall not be used where it may be necessary to disconnect and reconnect the circuit during servicing. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.40.14.10 Grounding: Ground connection to shields and to other mechanical parts, except the chassis or frame, shall not be made to complete electrical circuits, but only to eliminate high-potential points.

3.40.14.11 Wire and cable shielding: Shielding shall end at a sufficient distance from exposed conductors to insure against shorting or arcing between the conductor and the shielding.

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3.40.14.12 Identification of wiring: Wiring shall be identified by color-coding or, if approved by the contracting officer, may be identified by other means such as marking of terminals at both ends of each lead. Color-coding for chassis wiring shall be system 1 or 11 conforming to Standard MIL-STD-681, as applicable, except that coding in the model shall not be changed without permission of the contracting officer. Sleeve type cable marker, SM-C-603270-72, shall be secured on AC power cable for Power Supply PP-4763/GRC, using techniques applicable to heat shrinkable sleeving.

3.40.14.13 Crimping of solderless terminal lugs: Crimping of solderless terminal lugs shall be accomplished so that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of Specification MIL-T-7928.

3.40.15 Screws, other threaded devices, and related parts.

3.40.15.1 Form and fit: Unless otherwise specified, screw threads shall conform to Handbook H28, the Unified Thread Form and Series shall be class 2A or 2B fit, and the American National Thread Series shall be class 2 fit.

3.40.15.2 Self-tapping screws: Chips formed in "through-holes" by thread-cutting screws shall be removed.

3.40.15.3 Nuts, self-locking: Self-locking nuts shall conform to Specification MIL-N-25027. In no case shall locknuts containing plastic inserts be subjected to temperatures in excess of 250°F (as during baking of paint).

3.40.15.4 Thread engagement and extension: The length of thread engagement for tapped holes in steel or hard brass shall at least equal the nominal thickness of corresponding machine-screw nuts in sizes No. 12 and smaller and semifinished hexagon regular nuts in sizes 1/4 inch and larger, as shown in Handbook H28. Thread engagement for tapped holes in aluminum, magnesium, or other soft material shall be at least 50 percent greater than the thread engagement specified for steel or hard brass. Where permitted by design of the equipment, screws and bolts shall extend at least 1-1/2 threads beyond the nut, but no more than 1-1/2 threads plus 1/8 inch for screws up to 1 inch in length and 1-1/2 threads plus 1/4 inch for screws over 1 inch in length.

3.40.15.5 Torque-measuring devices: Threaded parts of steel shall be tightened to the torque values shown in table I. Threaded parts of other size or material shall be tightened to a suitable design value established by the contractor for the particular application. For critically stressed applications, the torque for tightening threaded parts shall be controlled by use of torque measuring or controlling devices.

Table I - Torque for tightening steel threaded parts

Thread	Torque (lb-in.)
6-32	7-9
8-32	14 to 18
10-24	19 to 23
10-32	25 to 30
12-24	32 to 39

3.40.15.6 Anti-seize compound: Anti-seize compound shall be used to prevent seizing of threads where the threaded surface is aluminum or magnesium alloy. The compound shall be made of equal parts by weight of petrolatum and zinc-dust-200 mesh-fine.

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3.40.15.7 Flat head machine screws: Flat head screw heads shall be completely seated in the material.

3.40.16 Tubes, electron; and semiconductors, transistor and diodes: Except as otherwise provided by Standards MIL-STD-200 and MIL-STD-701, these electron devices (tubes and semiconductors) shall conform to Specification MIL-E-1 and MIL-S-19500.

3.40.16.1 Choice, use and application: Except as otherwise specified in "a" and "b" below, the choice, use and application of these electron devices shall conform to Standards MIL-STD-200 and MIL-STD-701:

a. When substitution of a preferred or guidance type of electron device for a type required by the equipment specification would degrade equipment performance or ruggedness below specified requirements, pertinent data shall be furnished to the contracting officer, to justify retaining the type required by the equipment specifications.

b. When the foregoing standards require modification of equipment circuitry, the modification shall be subject to prior approval by the contracting officer.

3.40.16.2 Circuit location: The electron devices and non-linear parts furnished in equipment on contract shall be the individual devices used when the equipment was inspected for acceptance and, unless separate packaging of the devices is specified, shall remain in the respective locations occupied during the inspection.

3.40.17 Adjustment and repair: The equipment shall be so constructed that parts, terminals, wiring, etc. are accessible for circuit checking, adjustment, maintenance, repair, and replacement with minimum disturbance to other parts and wiring. After normal maintenance and repair, the equipment shall meet specified performance requirements.

3.40.18 Fabric and thread: The color of any required fabric shall be Olive-Drab No. 7. The color of thread shall match shade S-1 of the United States Army Standard Color Card for Official Standardization Shades of Sewing Thread (See 6.6). Fabric and thread shall be preshrunk, or allowance shall be made for shrinkage, in order to provide for satisfactory fit of finish items before and after they are immersed in water and then dried.

3.40.19 Interchangeability: Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment (See 4.15). Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.40.20 Controls and rotating components: Controls and rotating components shall conform to the following requirements:

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3.40.20.1 Effect of vibration, bounce, and shock: Controls shall maintain their setting when the items in which they are used are subjected to specified operating service conditions.

3.40.20.2 Mechanical operation: Play and backlash shall not cause poor contact or inaccurate setting. Controls shall operate freely and smoothly and shall be lubricated when lubrication does not interfere with operation.

3.40.20.3 Locking devices: Locking devices shall be so adjusted that they are capable of retaining controls in any given setting within the range of the control. The locking and unlocking action shall be easily accomplished and shall not affect the setting of the control. When in the unlocked position, the locking device shall not interfere with the normal operation of the control.

3.40.21 Printed wiring assemblies: Printed wiring assemblies shall be processed and fabricated according to Specification MIL-P-55110. Conformal coatings for the printed wiring assemblies shall be in accordance with Specification MIL-I-46058, except that the thickness of coating shall be .002 ± .001 inches.

3.41 Service conditions: The equipment shall meet the service conditions requirements as listed below, when tested in accordance with applicable paragraph of Section 4.

3.41.1 Temperature:

a. Operating: The equipment shall pass the test of 4.9.1 with ambient air temperatures surrounding the shelter, in the range -50°F to +120°F, with full impact of solar radiation at a rate of 360 BTU/ft ^{2}/hr, and ambient air temperatures within the shelter, in the range +32°F to +131°F (see 4.9.1).

b. Non-operating: Continuous exposure in the range -65°F to +155°F (see 4.9.1).

3.41.2 Relative humidity: The equipment, except Government furnished teletypewriters, teletypewriter paper, tape and ribbons, shall pass the test of 4.9.2 when subjected to relative humidity of 100% at all temperatures below +85°F, including conditions wherein condensation takes place in and on the equipment, and relative humidity corresponding to +85°F dew point at all temperatures between +85°F and +120°F and low relative humidity of 5% at temperature of +120°F.

3.41.3 Elevation: The equipment shall pass the tests of 4.9.3 to demonstrate compliance with the following altitude requirements:

a. Operating: Elevation up to 10,000 feet above sea level.

b. Non-operating: Elevation up to 50,000 feet above sea level.

3.41.4 Vibration, resonances: The amplitude of any contractor-furnished part, sub-assembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment component at any frequency between 10 and 55 cycles per second (cps). (See 4.9.4).

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3.41.5 Vibration, operating: The equipment components furnished by the contractor shall meet specified performance when subjected to vibration frequencies between 10 and 55 cps. Meeting the test of 4.9.5 shall be considered compliance with this requirement.

3.41.6 Bounce: Equipment components furnished by the contractor shall pass the test of 4.9.6 when subjected to $7\frac{1}{2}g \pm 2\frac{1}{2}g$ input, with random peaks in excess of 10g. Minor paint abrasion due to this test is permissible.

3.41.7 Shock: Bench Handling: Equipment components furnished by the contractor shall pass the test of 4.9.7 to demonstrate ability to withstand shock encountered during servicing and repair. Minor paint abrasion due to this test is permissible.

3.41.8 Rain: The AN/GRC-142( ), with shelter doors locked, and heater ports covered, shall show no evidence of water leakage into the shelter when exposed to precipitation at a variable rate on the exterior surface of the shelter, to a total of 9.50 inches over a 12 hour period, with intermittent wind speeds to a maximum of 35 knots. (see 4.9.8).

3.41.9 Fording: The Radio Teletypewriter Set in its shelter, with doors locked, shall be capable of being immersed in water to a depth of 21" measured from the bottom of the shelter skids, with no leakage or damage to the structure. The set shall be capable of meeting this performance requirement without the use of special fording kit. (See 4.9.9).

3.41.10 Vehicular transport.

3.41.10.1 S-318/G shelter configuration: The equipment shall be capable of being operated when transported on truck, cargo, 4X4, 3/4 ton, M-37 over cross country terrain. The test of 4.9.10.1 shall demonstrate compliance.

3.41.11 Orientation.

a. Operating: The AN/GRC-142( ) shelter, fully equipped and installed on its vehicle, shall be capable of operation when negotiating transverse grades up to 40% and longitudinal grades up to 60%. Compliance shall be checked in accordance with 4.9.10.2.

b. Non-operating: The AN/GRC-142( ) shelter, fully equipped, shall be capable of indefinite storage in its normal operating position. Approval of the first article samples shall be considered as compliance with this requirement.

3.41.12 Rail transport: The equipment including all components, GFE and CFE, shall be capable of being loaded, blocked and braced on a flat car, for shipment by rail and shall withstand the test of 4.9.10.5 without sustaining permanent deformation and shall meet specified performance after performing the test.

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3.41.13 Fungus: The equipment including all components, GFE and CFE, shall withstand, in both operating and non-operating conditions, exposure to fungus growth as encountered in tropical climates. Compliance shall be checked in accordance with 4.9.11. The contractor is not required to fungus-proof Government furnished items. The contractor is required to notify the Contracting Officer of any failure of GFE to pass the test of 4.9.11.

3.41.14 Salt fog: All contractor furnished components, fabricated parts and hardware shall be capable of withstanding the effects of salt fog as encountered in coastal areas. Compliance shall be checked in accordance with 4.9.12.

3.41.15 720 hour run: When tested as specified in 4.10.7, full compliance with specification limits is required during and after completion of the test. If one failure occurs within the first 72 hours, or if more than three failures occur during the remainder of the test, except for vacuum tube replacement, the equipment shall be rejected. The equipment shall also be rejected if a given failure occurs more than once, or if the single failure can be attributed to a design deficiency. A failure is defined as any change in equipment performance beyond specification limits. Any trip-off of a safety or protective device, or if the equipment unkeys for any reason other than that attributable to the line power input exceeding specification limits shall also be considered a failure. The daily tests and the tests at the end of the run shall meet specification requirements.

3.41.16 Ambient noise levels: When tested in accordance with 4.11, the sound pressure level in each octave band shall not exceed the level specified in Table II when measured at any microphone location where the operator's head may be located, during high speed operation of the main ventilating fan for ventilating the PP-4763/GRC. The octave band measurements shall be made using equipment in accordance with ASA Specification Z24.10-1953. Minor variations, not in excess of 2 db, in the octave bands are acceptable provided the arithmetic means of the four highest octave bands (600-1200; 1200-2400; 2400-4800; 4800-9600) does not exceed 55 db reference 0.0002 microbar. Every effort shall be made to mount the AN/GRC-106, teletypewriters, fans and heater so as to minimize acoustic noise.

Table II

Octave Band Frequency Limits (cycles/sec)	Sound Pressure Levels (db re 0.0002 microbar)
37.5 - 75	76
75 - 150	69
150 - 300	64
300 - 600	59
600 - 1200	57
1200 - 2400	55
2400 - 4800	53
4800 - 9600	52

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3.41.17 Air Transportability (Rotary Wing Aircraft): The shelter S-318/G with gross weight of 1,580 lbs. shall be capable of being picked up by a helicopter, or crane, transported while suspended from the helicopter or crane and lowered to any predetermined point on the ground. The shelter and its load shall be capable of withstanding 2.75G's acceleration with the shelter suspended by its lifting eyes. The test of 4.9.13 shall demonstrate compliance.

3.42 Preconditioning.

3.42.1 Bounce preconditioning: The components as listed below shall be capable of meeting the requirements specified herein without subsequent processing, after subjection to the bounce preconditioning of 4.8.1.

a. Switch Box Assembly SA-1554/GRC-142.

b. Power Distribution Panel SB-3018/GRC-142.

c. Switch Box SA-1555/GRC-142.

d. Interconnecting Box J-2728/GRC-142.

e. Remote Control C-7279/GRC-142.

3.42.2 Electrical preconditioning: The complete AN/GRC-142( ) shall meet test requirements of 4.8.2 during the "burn-in and debugging" test as called for therein.

3.43 Workmanship: The equipment shall be manufactured and assembled in accordance with applicable portions of paragraph 3.40 and other detailed requirements of this specification, and Specification MIL-P-11268. The contractor shall show in detail his approach to compliance with the "Zero Defects" concept insofar as workmanship is concerned, for the first article samples and all production models called for in the bid request and contract.

3.44 Running spare parts and instruction books.

3.44.1 Running spare parts: Running spare parts shall be provided as specified in the bid request and contract. Running spare parts shall be identical to corresponding intergal parts in the equipment. Running spare parts will be packaged in a manner that prevents damage when subjected to any service condition requirements of this specification.

3.44.2 Instruction books: Two sets of instruction books AN/GRC-122/142 shall be provided with each unit on order, as specified in the bid request and contract. See Military Specification MIL-M-10953.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection: Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection: Inspection shall be classified as follows:

a. First article inspection (does not include preparation for delivery. See 4.5).

b. Inspection covered by subsidiary documents. See 4.6.

c. Quality conformance inspections:

(1) Quality conformance inspection of equipment before preparation for delivery. See 4.7.

(2) Quality conformance inspection of preparation for delivery. See 4.16.

4.3 Test plan: A sectionalized test plan for the testing of the first article units and Groups A, B, C inspection sampling procedures, shall be submitted by the contractor to the COTR for his approval and shall be sufficiently comprehensive to insure compliance with all requirements specified herein. The test plan shall include a test for each requirement indicated in the 3 Section and any additional testing deemed necessary. The format for the plan covering Tempest and other classified types of tests, will be furnished to the successful bidder, upon request to the Contracting Officer's Technical Representative.

4.3.1 Test plan coverage: The test plan shall contain complete information on each test, including the following:

a. Title of test with identifying paragraph number in Sections 3 and 4.

b. Detailed description of all tests and test equipment.

c. Detailed description of test arrangements including all cables, fittings, and accessories required.

d. Location and identification of test points or connections applicable to each test.

e. Adaptability to automatic testing and equipment recommended.

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f. Method of test and complete procedure.

g. Data sheets containing the following:

(1) Title of test, with identifying paragraph number in Sections 3 and 4.

(2) Marked spaces for entry of test data.

(3) Test requirements and acceptable limits and tolerances.

(4) Marked spaces "Passed_____/ Failed_____".

(5) Marked spaces for dated signatures of person making the test and person witnessing the test.

(6) Date test started: _____19_____.

(7) Date test completed: _____19_____.

(8) Marked spaces for entry of information regarding test equipment, to include make, model number, serial number, calibration dates.

(9) Marked spaces for entry of serial number of item under test.

4.3.2 Tempest and secure radio teletypewriter operational tests: These test plans shall follow the same pattern as outlined in 4.3.1. However, these plans shall be prepared, protected and submitted in accordance with applicable Industrial Security regulations. The test plans for these tests shall be submitted as a separate document.

4.3.3 Electromagnetic Compatibility Test Plan: The test plan for electromagnetic compatibility shall be furnished as a separate section in accordance with paragraph 4.3 of MIL-STD-461.

4.4 Equipment compliance report for Tempest and EMC: The contractor shall submit an equipment compliance report, in accordance with Specification SCL-2101, with each first article sample that is submitted for acceptance. Compliance reports for TEMPEST and for electromagnetic interference shall be provided under separate cover. EMC reports shall be in accordance with paragraph 4.4 of MIL-STD-461. Compliance reports for TEMPEST tests shall be prepared, protected and submitted in accordance with applicable Industrial Security regulations and the DD-254 Form which covers contractual effort on the AN/GRC-142( ).

4.5 First Article inspection: This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the first article inspection specified in Table III; the inspection specified in the subsidiary documents covering the items listed in 4.6, and the inspection specified for Group A, Group B and Group C (see Tables IV, V and VI) respectively. The order of testing will depend on the number of first article samples specified in the contract and will be directed by the COTR.

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TABLE III - FIRST ARTICLE INSPECTION

INSPECTION	REQ.PARA.	INSP.PARA.
Temperature	3.41.1	4.9.1
Relative humidity	3.41.2	4.9.2
Elevation	3.41.3	4.9.3
Vibration, Resonances	3.41.4	4.9.4
Vibration, Operating	3.41.5	4.9.5
Bounce	3.41.6	4.9.6
Bench Handling	3.41.7	4.9.7
Rain	3.41.8	4.9.8
Deep Fording	3.41.9	4.9.9
Vehicular Transport	3.41.10	4.9.10.1
Orientation	3.41.11	4.9.10.2
Center of Gravity	3.5	4.9.10.3
Weight and Distribution	3.5	4.9.10.3, 4.9.10.4
Rail Transport	3.41.12	4.9.10.5
Fungus	3.41.13	4.9.11
Salt fog	3.41.14	4.9.12
Transmission continuity	3.9.6	4.10.6
EMI	3.9.9	4.10.10
TEMPEST	3.9.10	4.10.11
720 Hour Test	3.41.15	4.10.7
Air Transportability	3.41.17	4.9.13

4.6 Inspection covered by subsidiary documents: The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

ITEM	WHERE REQUIRED
* EMI	3.9.9
* Compliance to FED-STD -222	3.9.10
Cable Assemblies	3.9.11,
Finish	3.40.2
Marking	3.40.3

*This inspection shall be performed during first article testing only.

4.7 Quality conformance inspection of equipment before preparation for delivery: Each equipment produced under this specification shall be subjected to an electrical preconditioning run-in (see 3.42.2 and 4.8.2) prior to selection of samples for inspection. The contractor shall perform the inspections specified in 4.6 and 4.7.1 through 4.7.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government -- at its discretion -- may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment and personnel for Government verification inspection shall be made available by the contractor. Each component which will be subjected to Group A, B, or C inspection shall be bounce preconditioned (see 3.42.1) after final assembly, and prior to installation in the AN/GRC-142( ) configuration.

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4.7.1 Group A inspection: This inspection, including sampling, shall conform to Table IV and the inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order using these general inspection levels which is satisfactory to the Government.

TABLE IV - GROUP A INSPECTION

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual-Mechanical			1%	4%
TT-523	3.23	4.13	1%	4%
Switch Box Assy SA-1554	3.17	4.13	1%	4%
Power Dist.Panel SB-3018	3.18	4.13	1%	4%
Switch Box SA-1555	3.19	4.13	1%	4%
Interconnecting Box J-2728	3.20	4.13	1%	4%
Remote Control C-7279	3.29			
Bracketing	3.11.2	4.13	1%	4%
Cables	3.40.1	4.13	1%	4%
Shelter with Equipment	3.11) 3.43)	4.13)		
Safety & Protective Devices	3.9.8	4.13)	2.5%	6.5%
Physical and Mechanical Safety Measures	3.10	)		
Electrical		)		
Continuity		)	1.0%	
Functional Check		)		
On Switches and Wiring				
AN/GRC-106 (Operational)	3.9	4.14)		
Voice Operation of AN/GRC-106 and AN/GRA-6	3.28	4.8.2.1b(26)	1.0%	
Lock Out Relay and Local with AN/GRA-6	3.20.1e) 3.20.2e) 3.9.8.1)	4.8.2.1b(4))		
Remote to Local TA-312 and CW operation	3.6.2	4.8.2.1b(18)) 4.8.2.1b(27))		
Electrical System				
Preconditioning	3.42.2	4.8.2)		
Freq.Range	3.9.3	4.8.2.1b(14))	1.0%	
Power Output R.F.	3.9.4	4.10.5)		
Hum and Noise	3.9.5	4.10.3)		
Power Drain	3.9.2	4.10.9)		

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Inspection	Req Para	Insp Para	AQL
Electrical System (cont)			
Teletypewriter System	3.8.1	4.8.2.1b(16) (17) (21) (23) (24) (25) (27) (28) c.(17) d.	1%
Operational	3.6	4.14	1%

4.7.2 Group B inspection: This inspection, including sampling, shall conform to Table V and to the specification procedures for small sample inspection of MIL-STD-105, the reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

TABLE V - GROUP B INSPECTION

Inspection	Req Para	Insp Para	AQL
AN/GRC-142			
Water tightness	3.41.8	4.9.8	2.5% Level S-3
Interchangeability	3.40.19	4.15	6.5% Level S-3

4.7.2.1 Group B Sampling Plan: The acceptable level A.Q.L. shall be 6.5% and 2.5% defective as specified in Table V, and the inspection level shall be S-4 for normal and S-3 for reduced inspection.

4.7.2.2 Order of inspection within Group B: Group B inspection shall be performed in any order which is satisfactory to the Government.

4.7.3 Group C inspection: This inspection shall be listed in Table VI and shall be performed on sample units that have been subjected to and met Group A and B inspection (6.3.1).

TABLE VI - GROUP C INSPECTION

Inspection	Req Para	Test Para
AN/GRC-142		
Group I		
720 hour test	3.41.15	4.10.7
Group II		
Military Truck (Munson Course)	3.41.10	4.9.10.1
Air Transportability	3.41.17	4.9.13
Deep Fording	3.41.9	4.9.9

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4.7.3.1 Sampling for inspection of equipment: Units selected for Group C inspection shall be selected without regard to their quality in accordance with the following:

1 Unit from the first fifty (50) production units
(not including first article samples)

1 Unit from the remainder of the production units
(not including first article samples)

4.7.3.2 Noncompliance: If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements or if the contractor cannot determine the cause of failure, the matter will be referred to the contracting officer (See 6.3).

4.7.4 Reinspection of conforming group B and group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A reinspection after repair of all visible and known damage.

4.8 Preconditioning.

4.8.1 Bounce preconditioning (see 3.42.1): Each of the following components will be subjected to bounce preconditioning:

a. Switch Box Assembly SA-1554/GRC-142

b. Power Distribution Panel SB-3018/GRC-142

c. Switch Box SA-1555/GRC-142

d. Interconnecting Box J-2728/GRC-142

e. Remote Control C-7279/GRC-142

4.8.1.1 Procedure. The unit, with shock mounts (if any), removed or blocked, shall be placed in its normal operation position on the table of the Type 1000-SC Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned or otherwise changed prior to subjection to acceptance inspection.

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4.8.2 Electrical preconditioning (Burn-in and debugging) (See 3.42.2): This test shall precede other electrical tests and shall directly follow bounce preconditioning of components specified in 4.8.1, after those components are installed and/or stowed in the shelter. Each equipment on contract shall be subjected to this test. The test shall be performed under factory ambient conditions of temperature, relative humidity, and barometric pressure. A.C. and D.C. power input shall meet the criteria specified in 3.9.2. A.C. and D.C. voltages shall be adjusted to nominal values (28.5 volts D.C. and 115 volts AC).

4.8.2.1 Procedure: The following tests shall be performed, in the order given:

a. Continuity of all installed wiring and cabling including that provided for full duplex operation.

b. D.C. primary power operation (AC/DC Power Transfer switch (3.9.2c) in "DC" position and AC circuit breakers (3.11.3.6) OFF).

(1) Lights and interlock switch operation.

(2) Power Panel D.C. voltmeter and lamp indicator operation.

(3) Main ventilating fan operation on full speed and half speed.

(4) Lockout circuit operation and override control.

(5) Inverter operation, and proper AC output voltage.

(6) Teletypewriter motor operation and light operation with inverter on.

a Main TT-98/FG

b TT-76B/GGC (or TT-76C/GGC)

(7) Voltage with proper polarity, on main security device D.C. power cable connector.

(8) Voltage, with proper polarity, on auxiliary security device D.C. power cable connector.

(9) Voltage, with proper polarity, on auxiliary inverter D.C. power cable receptacle.

(10) Voltage, with proper polarity, on 28 VDC convenience outlet receptacle.

(11) Multi-fuel heater operation (simulated if required by factory safety regulations).

(12) Voltage, with proper polarity, on auxiliary RT-662/GRC power cable connector.

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(13) Voltage, with proper polarity, on auxiliary RT-662/GRC R.F. Preselector power cable connector, if cable is required in bid request and contract.

(14) AN/GRC-106 operation in receive mode at five spot frequencies in the range 2.0 to 29.9 MHz.

(15) MD-522A/GRC operation in all modes, with AN/GRC-106 in receive mode, on one or more of the spot frequencies of 4.8.2.1b(14) above.

(16) Radio teletypewriter reception on an appropriate operating frequency, to check compliance with requirements of 3.8.1 regarding operating capabilities.

(17) Reception of voice and radio teletypewriter test messages transmitted to the r.f. input connector of the RT-662/GRC, with a signal level equivalent to one (1) microvolt, to check compliance with requirements of 3.6 regarding quality of voice and radio teletypewriter copy in all operating modes provided by Modem MD-522A/GRC, in OWR service.

(18) Two-way order wire operation using local and remote telephones, type TA-312/PT, per 3.6.2.

(19) Two-way order wire operation, using local and remote units of AN/GRA-6, per 3.6.2.

(20) Remote voice reception, using local and remote units of AN/GRA-6, with RT-662/GRC tuned to an appropriate frequency.

(21) Remote voice and radio teletypewriter reception of signals transmitted to the shelter, per 4.8.2.1b(17) above, to check compliance with requirements of 3.6 as described in 4.8.2.1b(17) above.

(22) AN/GRC-106 local operation in the transmit mode, into a 50 ohm R.F. dummy load, at two spot frequencies in the range 2.0 to 29.9 MHz, using the following types of modulation:

a SSB voice, using M-29/U microphone.

b AM voice, using M-29/U microphone.

c CW, using KY-116/U.

(23) AN/GRC-106 and MD-522A/GRC local operation in the transmit mode, into a 50 ohm R.F. dummy load, at two spot frequencies in the range 2.0 to 29.9 MHz, in all operating modes provided by MD-522A/GRC for OWR service.

(24) Radio teletypewriter transmission to a test station receiving facility, at one spot frequency in the range 2.0 to 29.9 MHz, to check compliance with requirements of 3.8.1 regarding operating capabilities.

(25) Transmission of voice and radio teletypewriter test messages to a test station receiving facility at one spot frequency in the range 2.0 to 29.9 MHz to check compliance with requirements of 3.6 regarding quality of voice and radio teletypewriter copy in all operating modes provide by Modem MD-522A/GRC, in OWR service.

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(13) MD-522A/GRC operation in all modes, with AN/GRC-106 in receive mode on appropriate spot frequencies.

(14) AN/GRC-106 and MD-522A/GRC local operation in the transmit mode, into a 50 ohm R.F. dummy load, at one spot frequency in the range 2.0 to 29.9 MHz, in all operating modes provided by MD-522A/GRC, for OWR service.

(15) Transmission of voice and radio teletypewriter test messages to a test station receiving facility at one spot frequency in the range 2.0 to 29.9 MHz, to check compliance with requirements of 3.6 regarding quality of voice and radio teletypewriter copy in all operating modes provided by Modem MD-522A/GRC, in OWR service.

d. Continuous "key down" transmission in 850 Hz FSK mode, with AN/GRC-106 operating into a 50 ohm R.F. dummy load at one spot frequency in the range 2.0 to 29.9 MHz, and tape copy being transmitted through TT-76B/GGC (or TT-76C/GGC) with home page copy being printed on TT-98/FG. This test shall be run, without interruption, for a period of six hours. Spot checks at least once every half hour shall be made to determine compliance with requirements of 3.8.1c, 3.9.4.3 and 3.9.5. This test shall be run when operating from AC primary power, with A.C. input voltage adjusted to 115 volts ± 5 volts. Failure of any component, due to contractor-furnished equipment malfunction and/or workmanship, or failure of any contractor-furnished component during these electrical preconditioning tests shall be cause for rejection.

4.9 Service conditions tests (see 3.41).

4.9.1 Temperature test (see 3.41.1): The radio teletypewriter set shall be subjected to the temperature cycle shown on Standard MIL-STD-169, except that Step 2 shall be +155°F, Step 4 shall be +120°F, Step 6 shall be -65°F, and Step 8 shall be -50°F. In addition, thermocouples shall be placed at all locations where high temperature is expected.

a. Energize AN/GRC-142 for Step 1 of temperature test. Turn main ventilating fan ON. Open door intake louvre. Leave multi-fuel heater OFF.

b. Make the following measurements at Step 1:

MEASUREMENT	TEST PARA.	REQ. PARA.
Operational Check	4.8.2.1c(15)	3.6 (A.C. primary power)
R.F. Power Output	4.10.5	3.9.4 (D.C. primary power)

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(26) Remote voice transmission, using local and remote units of AN/GRA-6, with AN/GRC-106 transmitting to the test station receiving facility, per 4.8.2.1.b(25) above, to check voice quality.

(27) Remote CW transmission using Key KY-116A/U at the remote site, with AN/GRC-106 transmitting to the test station receiving facility per 4.8.2.1. b(25) above, to check compliance with requirement of 3.9.7.

(28) Remote radio teletypewriter transmission, using Remote Control C-7279/GRC-142 and local and remote units of AN/GRA-6, with AN/GRC-106 transmitting to the test station receiving facility, per 4.8.2.1.b(25) above, to check quality of radio teletypewriter copy.

c. A.C.Primary power operation (AC/DC Power Transfer Switch (3.9.2c) in "AC" position, and main DC circuit breaker (3.9.8.3c) OFF.

(1) PP-4763/GRC operation.

(2) Lights and interlock switch operation.

(3) A.C. voltmeter operation.

(4) Main ventilating fan operation on full speed and half speed.

(5) Lockout circuit operation and override control.

(6) Teletypewriter motor operation.

a. Main TT-98/FG.

b. TT-76B/GGC (or TT-76C/GGC).

(7) A.C. voltage on all A.C. outlet receptacles.

(8) A.C. voltage on main security device A.C. power cable connector.

(9) A.C. voltage on auxiliary security device A.C. power cable connector.

(10) Voltage, with proper polarity, on auxiliary RT-662/GRC power cable connector.

(11) Voltage, with proper polarity, on auxiliary RT-662/GRC R.F. Preselector power cable connector, if cable is required in bid request and contract.

(12) AN/GRC-106 operation in receive mode at one spot frequency in the range 2.0 to 29.9 MHz.

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c. Turn off AN/GRC-142( ) and close shelter door and intake louvre.

d. On reaching Step 4, open the door intake louvre, turn on main ventilating fan (full speed) and operate the AN/GRC-142( ) in transmit mode using the dummy whip antenna, 4.10.1.4 on a radio frequency to be specified by the COTR, as being the frequency which produces greatest heat dissipation in the AM-3349/GRC-106. At this frequency, perform the measurements as listed in "b" above. Next, set the AN/GRC-142( ), on A.C. primary power, to transmit continuously on 85 Hz NSK RTTY, with teletypewriters operating in accordance with 4.8.2.1d, for a period of four (4) hours on the radio frequency and dummy whip antenna mentioned above. Make spot checks and obtain performance as stated in 4.8.2.1d, during and at the end of the four hour key-down test. At the end of the four hour key-down test, perform the measurements as listed in "b" above.

e. On reaching Step 5, perform the measurements as listed in "b" above.

f. Turn off AN/GRC-142( ). Place an electric heater with a 15,000 BTU capacity in the shelter, at a location near the multi-fuel heater. Test the electric heater for operation. Turn the heater off. Close the shelter door, and door intake louvre.

g. On reaching Step 8, energize the AN/GRC-142( ) and turn on the electric heater used to simulate the multi-fuel heater. Leave shelter door and intake louvre closed. Leave main ventilating fan off. Turn on all other equipment, using A.C. primary power. Set AN/GRC-106 and MD-522A/GRC for SSB voice operation. Make SSB voice operational check per 4.8.2.1c(15) within ten (10) minutes after energizing the AN/GRC-142( ). Monitor temperature at base of operator's chair. When this temperature reaches +32°F, perform the measurements as listed in "b" above. Control the electric heater to maintain a shelter internal ambient temperature in the range +32°F to +70°F during the tests at Step 8, where teletypewriter operation is involved.

h. Turn off electric heater, and remove it from the shelter. Leave AN/GRC-106 operating in "standby" mode.

i. On reaching Step 10, open shelter door; turn on main ventilating fan and perform the measurements as listed in "b" above.

4.9.2 Relative humidity test (see 3.41.2).

4.9.2.1 Chamber: The chamber and accessories shall be constructed and arranged in such a manner as to avoid condensate dripping on the test item. The chamber shall be vented to the atmosphere to prevent the buildup of total pressure. Relative humidity shall be determined from the dry-bulb wet-bulb thermometer comparison method. The air velocity flowing across the wet-bulb shall not be less than 900 feet per minute. Provisions shall be made for

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controlling the flow of air throughout the internal test chamber area where the velocity of air shall not exceed 150 feet per minute. Distilled, demineralized, or deionized water at 25 degrees C (77 degrees F) shall be used to obtain the specified humidity. Prior to starting the test, the pH of the water shall be measured. The pH of the water shall not be less than 6.5 or greater than 7.5.

4.9.2.2 Procedure: The test item shall be placed in the test chamber in a manner simulating service usage. Dry the equipment at 130 ± 5 degrees F for 24 hours. Condition the equipment at 77 ± 5°F and 40 to 60 percent relative humidity for 24 hours. Take initial measurements in accordance with 4.9.2.3. (The equipment may be readjusted or realigned as necessary to meet specification requirements). No further realignment or readjustment shall be permitted other than with accessible controls employed for operation of the equipment. No repair or replacement of parts shall be permitted. Subject the equipment to five (5) continuous 48-hour cycles in accordance with Specification MIL-STD-170. During the last five (5) hours of each cycle, take measurements as specified in 4.9.2.3, with shelter door closed; shelter door intake louvre open; main ventilating fan on (full speed); multi-fuel heater off; and teletypewriter equipment remotely operated, outside of the chamber.

4.9.2.3 Performance: Measurements made during test and specified performance during cycling shall be as follows:

MEASUREMENT	TEST PARA.	REQ. PARA.
Operational Check	4.8.2.1c(15)	3.6 (A.C. primary power)
R.F. Power Output	4.10.5	3.9.4 (D.C. primary power)

There shall be no mechanical deterioration, as evidenced by corrosion of metal parts or binding of rotating parts.

4.9.2.4 Failure: If any equipment fails to meet the performance specified in 4.9.2.3 during cycling, it shall be realined or readjusted once. If the equipment then fails to meet 4.9.2.3 or fails subsequently during cycling, it does not pass the test. Teletypewriters, teletypewriter paper, tape and ribbons in the shelter, are exempt from this requirement. Defects in these items shall be noted and recorded in the test data.

4.9.3 Elevation test (see 3.41.3): The AN/GRC-142( ) shall be placed in an altitude chamber capable of simulating altitudes of 50,000 feet above sea level, and tested as follows:

a. At standard test conditions, 4.10.1, energize the AN/GRC-142( ), using A.C. primary power, per 3.9.2.b. Connect dummy whip antenna, 4.10.1.4, for 2.0 MHz to AN/GRC-106 whip antenna lead-in wire. Ground the dummy whip antenna. Perform the following measurements at 2.0 MHz:

MEASUREMENT	TEST PARA.	REQ. PARA.
Operational check (SSB voice only)	4.8.2.1c(15)	3.6 (A.C. primary power)
R.F. Power Output (SSB voice @ 2MHz only)	4.10.5	3.9.4 (D.C. primary power)

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b. Reduce the barometric pressure to 20.6 ± 0.1 inches of mercury (simulated 10,000 feet above sea level), and after temperature and pressure stabilization for 2 hours, repeat the measurements specified in 4.9.3(a). The equipment shall meet specified performance for those measurements.

c. Remove the power from the equipment and further reduce the barometric pressure to 3.4 ± 0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 2 hours.

d. Return the chamber to 29.9 ± 0.1 inches of mercury. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.9.3(a). The equipment shall meet specified performance for these measurements.

4.9.4 Test for internal vibration (see 3.41.4): Internal vibration of the equipments listed in 4.8.1 shall be measured as follows to determine conformance to 3.41.4:

a. Secure each of the equipments directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Attachment points shall simulate chassis securing points. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

b. Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cps. The total excursion of the applied vibration shall be at least .030".

c. In each of three directions, change the frequency in steps of one cps and maintain each frequency for at least 10 seconds.

d. Measure vibration amplitudes by optical means, or by other means, provided that vibration of the part is not affected by the measurement.

4.9.5 Vibration test - 90 minutes (see 3.41.5): The equipments listed in 4.8.1 shall be tested as follows to determine conformance to 3.41.5:

a. Secure the equipment in a test fixture, which simulates actual mounting method. This fixture shall be secured to a vibration table that can be controlled within 10 percent of the specified amplitude. The equipment shall be mounted in its normal operating position with connections (such as cable assemblies) in place.

b. Vibrate the complete equipment assembly successively in three mutually perpendicular directions over a frequency range of 10 to 55 cps. The total excursion of the applied vibration shall be 0.040 inch in the horizontal plane and 0.060 inch in the vertical plane.

c. In each of three directions, change the frequency in steps of one cps.

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d. The equipment shall be subjected to 30 minutes of vibration in each of three mutually perpendicular planes parallel to the equipment surfaces. Electrical tests in accordance with the test plan shall be made before the test, while being vibrated, and after the tests.

4.9.6 Bounce test (see 3.41.6). Each of the equipments, in turn, listed in 4.8.1 shall be bounce tested.

4.9.6.1 Test conditions.

a. Use a package tester, for testing military equipment, of suitable capacity as made by L.A.B. Corporation, Skaneateles, New York, or equal.

b. Cover the test bed of the package tester with a panel of ½ inch plywood, with the grain parallel to the drive chain. Secure the plywood with six penny nails, with top of heads flush with or slightly below the surface. Space nails at six inch intervals around all four edges. If the distance between either pair of fences is greater than 24 inches, the plywood shall also be nailed at three-inch intervals in a six-inch square at the center of the test area.

c. Using suitable wooden fences, constrain the vehicular, or simulated, adapter plate to a horizontal motion of not more than 2 inches in any lateral direction. Fences shall be a distance from the test item more than sufficient to insure that the equipment will not rebound from fence-to-fence. Additional barriers may be used to safeguard personnel. Reference Electronics Command Drawing SC-A-46439.

4.9.6.2 Procedure.

a. Establish compliance with the operational requirements under ambient environmental conditions, by following procedures contained in the test plan.

b. Secure the equipment in its test fixture which is secured to the vehicular, or simulated adapter plate, in its normal operating position and place on the package tester within the constraints outlined in 4.9.6.1c above.

c. Attach an accelerometer, as close as possible to the point of equipment attachment, to record the shock transmitted to the equipment.

d. Adjust the package tester, shafts in phase and table operating in a vertical motion, to a speed such that the value of the random acceleration peaks shall be 7.5 ± 2.5 g's. Measure this input with an accurate measuring or recording system incorporating a 100 cycles per second low pass filter. Due to the random nature of the input, pulses greater than 10 g's can be expected to occur, however, if they are infrequent, they need not be used in calculating the average. In no case shall the speed of the package tester exceed 285 revolutions per minute. Conduct the test for a total of three hours. At the end of each 3/4-hour period, rotate the adapter plate and test item 90 degrees, each time in the same direction.

e. Before, after, and during each phase the tests listed in the test plan shall be made on the equipment, and specified performance shall be obtained, without need for adjustment or replacement of parts.

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4.9.7 Bench handling test (see 3.41.7): Each equipment listed in 4.8.1 shall be placed in a suitable position for servicing on a horizontal solid wooden bench top at least two (2) inches thick. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing.

Step a. Using one edge as a pivot, lift the opposite edge of the chassis until one of the following conditions occurs (whichever occurs first), then let the chassis drop back freely to the horizontal bench top. Repeat, using other practical edges of the same horizontal face as axes, for a total of four drops.

(1) The chassis forms an angle of 45 degrees with the test bench top.

(2) The lifted edge of the chassis has been raised four (4) inches above the test bench top.

(3) The lifted edge of the chassis is just below the point of perfect balance.

Step b. Repeat step a, with the assembly resting on other faces until it has been dropped for a total of four (4) times on each face on which the assembly could be placed practicably during servicing. The test item shall not be operating during the test. At the conclusion of the test, the test item shall be operated and tested in accordance with the test plan, and specified performance shall be obtained, without need for adjustment or replacement of parts.

4.9.8 Rain Test (see 3.41.8): The AN/GRC-142( ), with shelter doors locked and heater ports covered, shall be tested in accordance with Method 506 of MIL-STD-810. The wind shall be maintained for 15 minutes during each 30 minute test period. At the conclusion of the test, there shall be no evidence of leakage of water into the shelter. The interior of AC and DC entrance panels shall be examined during or after the rain test.

4.9.9 Deep fording test (see 3.41.9): The equipment, prepared for transportation on a military cargo truck, shall be immersed in water to a depth of 21 inches measured from the bottom of the shelter skids. No special fording kits shall be used in the performance of this test. No additional caulking or taping is permitted after submission of sample. The equipment shall be immersed for 1 hour, and shall be restrained by external tie-downs. There shall be no evidence of leakage into the shelter or into the walls, floor or door. To determine that no water has entered space between the skins, holes shall be made at points to be determined by the Government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.9.10 Mechanical tests: The following tests will be performed at facilities located at Aberdeen Proving Grounds, Aberdeen, Maryland. These facilities will be made available to the contractor at a time and cost to be determined by Aberdeen Proving Grounds. One set will be subjected to these tests, another set shall be used as a monitoring station while these tests are being performed.

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4.9.10.1 Munson Course (see 3.41.10.1 and 3.9.6): One of the complete installations ready for field use and operating in accordance with 4.8.2.1b(14), (15), (16), (24) and (25) shall be driven ten times around the Munson Course. There shall be no evidence of equipment degradation during of after completion of this test. Errors in teletypewriter copy during operation in motion on the Munson Course shall be analyzed to determine whether the errors are due to loose parts and/or wiring. If so, the equipment shall be considered as failing the test. If errors are found to be due to design characteristics of the Government-furnished teletypewriters, then this deficiency shall not be counted as a failure in the AN/GRC-142( ). The following sections of the Munson Course shall be used and these sections shall be traveled at the speed indicated:

a.	Coarse Washboard	5 mph
b.	Belgian Block	20 mph
c.	Radial Washboard	15 mph
d.	Single corrugations	20 mph
e.	Any short sections between above sections	20 mph

4.9.10.2 Orientation test (see 3.41.11): The complete installation, ready for field use and operating shall be subjected to grades of 60% for-and-aft and 40% laterally. The equipment shall be capable of operating in accordance with 4.8.2.1b(14), (15), (16), (24) and (25) during this test with no degradation.

4.9.10.3 Center of gravity: The center of gravity of the complete installation shall be determined (see 3.5).

4.9.10.4 Total weight and distribution: The total weight and distribution of the complete installation will be determined (see 3.5).

4.9.10.5 Rail transport impact test (see 3.41.12).

4.9.10.5.1 Test conditions:

a. The equipment shall be mounted on a rail-road flat car in accordance with Figure 1 of MIL-S-55429 The cabling and blocking holding the shelter in place shall be tight at the start of each impact run.

b. Four (4) railroad cars are required, one (1) car containing the equipment to be tested, one (1) car loaded to 165,000 pounds gross weight and two empty cars. All cars shall be equipped with American Association Railroads standard draft gear. The test shall be conducted on a flat section of track.

c. The car containing the equipment shall be positioned ahead of the two (2) empty cars, all of which shall be coupled together with their brakes off.

4.9.10.5.2 Test method.

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a. Establish compliance with operational requirements under prevailing ambient environmental conditions, by performing tests as listed in 4.8.2.1c (12), (13) and (15).

b. The car loaded with the 165,000 pound gross weight shall be set in motion to a speed of 9 ± .5 miles per hour and impacted against the car containin the equipment. Four impacts shall be performed as follows: With the shelter positioned longitudinally on the flat car, apply two impacts to each end of the equipment car.

c. If blocking or tie downs holding the equipment to the test car are torn loose or broken by the impact, the test shall be repeated. A damage survey shall be made after each impact and the equipment shall have sustained no permanent deformation. Upon completion of this test, the equipment shall meet the operational requirements under prevailing ambient environmental conditions without effect on the performance determined in "a" above.

4.9.11 Fungus test (see 3.41.13): The equipment, including all Government and contractor-furnished components, shall be tested in accordance with MIL-STD-810, Method 508.

4.9.12 Salt fog (see 3.41.14): All contractor-furnished components, fabricated parts and hardware shall be tested in accordance with Method 509, Procedure I of MIL-STD-810.

4.9.13 Air Transportability (see 3.41.17): The equipment shall have sustained no permanent deformation. Upon completion of this test, the equipment shall meet the operational requirements under prevailing ambient environmental conditions without degradation of performance.

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4.10 Electrical tests.

4.10.1 Standard test conditions. Unless otherwise specified, the following standard test conditions shall apply:

4.10.1.1 Atmospheres. All tests shall be made at prevailing factory conditions of temperature, humidity, and pressure, unless otherwise specified in the detailed procedure.

4.10.1.2 115 VAC and 28 VDC. The power source shall be 115 ± 2 volts alternating current, 60 Hz, and/or 28.0 ± 0.5 volts, direct current, except where specifically changed in the detailed procedure. The regulation shall be such as to maintain this voltage when the load is changed from no load to the maximum for any of the equipment under test. Noise and ripple on the dc voltage source shall be less than 0.5 volts, ac. The voltage sources shall be adjustable over the range of 24 to 30 volts dc and 103.5 to 126.5 volts A.C.

4.10.1.3 Dummy antenna. A 50-ohm resistive load shall be used except where the detailed procedure requires the use of a whip antenna. The 50-ohm antenna shall be non-reactive, and shall have a continuous duty rating of 500 watts.

4.10.1.4 Whip antenna. Where the detailed testing procedure requires a whip antenna, a dummy whip antenna shall be used unless otherwise specified herein. The dummy antenna characteristics must conform to Drawing SC-C-4500 and the characteristics as measured shall be submitted to the contracting officer's technical representative for approval.

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4.10.1.5 Grounding: A single ground connection shall be provided which shall connect to the internal grounding system of the equipment. Test equipment shall be grounded to the common equipment ground wherever practicable. The cases of metal cased test equipment shall be grounded and so noted in test record.

4.10.1.6 Standard voice test message: The contractor will furnish a magnetic tape upon which is recorded a test message to be used where specified in the detailed test procedure. The test message will consist of the following phrase repeated over and over: "Jo took father's shoe bench out, she was waiting at my lawn". The test message signal level shall be an average 20 MV (RMS)/600 ohms.

4.10.1.7 Warm-up period: Unless otherwise specified, measurements shall be made, after five minutes from the time primary power is applied.

4.10.1.8 Standard two-tone input: Unless otherwise specified in the detailed test procedure, the modulation applied to the equipment shall be two audio tones equal in amplitude (200 MV (RMS)/600 ohms) and whose frequencies shall be 1500 and 2500 cps.

4.10.2 Safety and protective devices: A series of tests shall be conducted to determine compliance with the specification relative to provisions for safety to equipment and personnel. All parts not provided with automatic protective devices in the appropriate circuits shall be inspected to determine that adequate protection is provided, including blackout switch operation, 3.11.3.10.7. Circuit breakers shall be tested to determine that they are resettable within a reasonable time after the overload has been removed. Lightning protective devices shall be tested - see 3.9.8.

4.10.3 Hum and Noise test: A suitable test shall be made to check for hum and noise in the output of the equipment. The requirements of 3.9.5 shall be met.

4.10.4 CW keying tests (see 3.9.7): CW keying shall be at 25 dot cycles per second (60 wpm). The rf output wave shape shall be checked for conformance to the requirements of 3.9.7.

4.10.5 RF power output test (see 3.9.4): The rf power output shall be measured, unless otherwise specified, using the standard conditions of 4.10.1. All rf power output measurements shall be made as a function of the voltage across the resistive element of the dummy antenna, and shall be corrected for temperature and frequency characteristics. Correction curves for frequency effects shall be furnished by the contractor for all rf meters used on these tests. Power output shall be measured at a total of ten (10) operating frequencies, as specified by the COTR. The dummy loads, 4.10.1.3 and 4.10.1.4 for each operating frequency shall be specified by the COTR.

4.10.6 Transmission continuity test (see 3.9.6): During the road test, 4.9.10.1, the output of the transmitter shall be monitored by a receiver which shall have an oscilloscope connected to its audio output. There shall be no evidence of momentary change or loss of signal due to an intermittent condition in the equipment under test.

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4.10.7 720 hour test (see 3.41.15): The AN/GRC-142( ) system shall be operated in a continuous transmitter "key-down" condition for a period of 720 hours at an external shelter ambient of +110°F ± 3°F. This test shall be performed using A.C. primary power, per 3.9.2b. The equipment shall operate in one way reversible radio teletypewriter mode, using 85 Hz narrow shift signalling. Teletypewriter equipment shall be switched to operate in accordance with 3.8.1d. The AN/GRC-106 r.f. power output shall be measured at the dummy whip load of 4.10.1.4 at a frequency of 2.0 MHz. The teletype machines shall be operated at 60 wpm. If excessive teletype errors are caused by GFE, the test shall be stopped, equipment repaired, and the test continued. Excessive errors caused by contractor equipment shall be cause for rejection. The requirements of 3.41.15 shall be met. The main ventilating fan shall be on during the complete test.

4.10.8 Metering tests: All meters on contractor-furnished items shall be checked for accuracy as a component, and for proper reading when installed in the system.

4.10.9 Power drain tests: Primary power input shall be measured and recorded under operating conditions specified in 4.8.2.1b(25) and 4.8.2.1c(15).

4.10.10 EMI Tests (see 3.9.9): The first article sample shall be tested in accordance with MIL-STD-462. The following tests shall be performed using D.C. primary power (3.9.2a) and repeated using A.C. primary power (3.9.2b) to show compliance with MIL-STD-461:

CE02	RE03	CS02
CE03	RE02	RS03 (Rerun RS03 in the frequency range of 20 to 200 MHz with the field intensity at 1CV/meter)
CE04	CS01	

The following tests shall be performed for informational purposes:

CE01	RE01	CS06

4.10.11 Federal Standard ...222 compliance (see 3.9.10): These tests shall be performed in accordance with the approved test plan (see 4.3.2 and 6.9).

4.11 Acoustic noise test (see 3.41.16): Using a suitable calibrated sound level meter, with a flat (non-weighted) response, acoustic noise shall be checked under conditions as given in 3.41.16. The requirements of 3.41.16 shall be met.

4.12 End stop test: A torque of 72 pound-inches shall be applied so as to check control end stops on contractor-furnished items. This torque shall be applied alternately 500 times in each direction (see 3.40.5).

with a message to

shall be notified immediately via telephone

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4.13 Visual and mechanical inspection: Visual and mechanical inspection shall be in accordance with MIL-STD-252 and the following inspection: Parts, components and complete equipments shall be inspected for workmanship, mechanical construction, loose nuts, screws, and other type fasteners, application of specified treatments and miscellaneous defects. The equipment and controls shall be inspected for proper mechanical and electrical operation. Wiring, soldered connections, grounds, welds, finishes, etc. shall be inspected for workmanship. Clearances, dimensions, and mechanical adjustments, including mounting provisions for security devices, shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles, when facsimiles are required. Examination also may be made for other visual or mechanical defects, similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified by the Government as major or minor in accordance with the definitions of MIL-STD-105. Also, check full, free movement of TT-98/FG and TT-76C/GGC on slides, after A.C. power cords are plugged into receptacles and cords are banded.

4.14 Operational system test: An "on the air" test using authorized frequencies, shall be conducted by transmitting and receiving radio teletypewriter, voice and CW signals between two Radio Teletypewriter Sets AN/GRC-142( ), situated a minimum of 200 feet apart. In this test, the equipments shall be operated on A.C. and D.C. primary power, in one way reversible and full duplex modes. Tests shall be made to determine compliance with the following:

a. The components shall provide the functions specified in Section 3.6.

b. The radio teletypewriter set shall meet the requirements of 3.6 and 3.8.

c. All components shall be properly grounded, and no undesirable RF currents shall be set up during transmitter operation.

d. The radio teletypewriter set shall provide a maximum of one error in each consecutive 20 lines of transmitted and received teletype copy.

For tests in remote and full duplex operation, auxiliary equipment, furnished by the Government, will be used. Doublet antennas will be erected end-to-end, at each AN/GRC-142( ) station, with a minimum of 150 feet separation between transmitting and receiving doublet antennas. Frequency separation of 10% shall be maintained between transmitting and receiving frequencies, during full duplex operation.

4.15 Inspection for interchangeability: The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.40.19. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, insertable assemblies, and replaceable parts, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases covers, and mountings.

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d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

4.16 Quality conformance inspection of preparation for delivery: Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5. Approval of method for packing, crating and transporting shall be obtained from the Contracting Officer, prior to any deliveries of equipment.

4.17 Inspection prior to shipping: All equipments produced on the contract will be subjected to the following inspection before packaging and shipping: 4.8.1, 4.8.2, 4.10.5, 4.13 and 4.14.

5. PREPARATION FOR DELIVERY

5.1 The equipment shall be prepared for delivery as specified in the contract.

5.2 MIL-S-55505 - Shelter Electrical Equipment (containing equipment). Packaging and Packing of.

6. NOTES

6.1 Ordering data: Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level of packaging and level of packing required for shipment. (Level A, Level B, or Level C).

c. The specific paragraphs of Section 5 which are applicable to this procurement.

d. First article inspection:

(1) Three sample units of AN/GRC-142( ) are generally required so that lengthy environmental tests can be completed on one sample unit while complete performance measurements can be made on the second and third sample unit (see 3.3).

(2) First article pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

e. Marking and shipping of samples.

f. Place of final inspection.

g. Technical literature required.

h. Quantity of tools and running spare parts required.

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1. Submission of the statement of treatment covering tropicalization of material MIL-P-11268 , as soon as possible after award of contract. This statement should be submitted to the Contracting Officer.

6.2 Definitions.

6.2.1 Equipment divisions. The terms used herein for equipment divisions conform to Standard MIL-STD-280.

6.2.2 Definitions of inspection terms: The inspection terms used herein conform to Standard MIL-STD-109. The following are abbreviated definitions selected from those given therein for immediate reference:

6.2.2.1 Inspection: Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.2.2.2 Examination: Examination consists of simple, generally non-destructive determinations of compliance, without use of special testing equipment.

6.2.2.3 Testing: Testing consists of determinations of compliance, using technical means.

6.3 Group C inspection: Approval to ship may be withheld, at the discretion of the Government, pending the decision from the Contracting Officer on the adequacy of corrective action. See 4.7.3.2.

6.3.1 Initial shipments: Initial shipments, of end items will be contingent only upon successful completion of all Group C tests. (4.7.3).

6.4 Location of operational inspection: It is desirable that the operational system test (4.14) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the operational test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.5 Nomenclature: The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/GRC-142W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Color of textiles: Standard samples of Olive Drab No. 7 may be obtained from the Sample Loan Unit, Military Clothing and Textile Supply, Philadelphia, Pennsylvania. When requesting samples, indication should be given as to type of material concerned.

6.7 Verification inspection: Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract.. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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6.8 Classification and clearance.

6.8.1 General: In general, the security clearance for the facility and personnel concerned, shall be cleared for SECRET. Special requirements apply to certain aspects of this work. See 4.3.2, and applicable DD-254 Form.

6.8.2 Specific.

6.8.2.1 The contractor's facility shall be approved for SECRET.

6.8.2.2 Personnel working on this contract shall be cleared for SECRET.

6.8.2.3 The contractor shall not divulge any information concerning any aspect of the work performed under the contract to any person not directly engaged in work under the contract, or not specifically designated by the contracting officer in writing to receive such information.

6.9 TEMPEST Tests.

a. Complete Tempest tests shall be performed by the contractor. It is to be understood that upon completion and evaluation of these tests, the Government will then determine which of the tests will be performed by direction of the contracting officer, in accordance with the "Changes" Article, on a sampling basis during production, to ascertain continued adherence to the specified requirements of Federal Standard 222.

b. Evaluation, by the Government, of the first article data resulting from the contractor-performed Tempest tests, will be the basis for first article approval of the Tempest requirements.

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RADIO TELETYPEWRITER SET AN/GRC-142( ) DATED 8 FEBRUARY 1968

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION
MIL-R-55587(EL) Radio Teletypewriter Set AN/GRC-112() in S-318 Shelter

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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